The stress function and the partial derivatives can be rewritten in terms of polar coordinates, by introducing $\rho^2 = x^2 + y^2$:

$$\Phi = xh\left[1 - \left(\frac{r}{\rho}\right)^2\right] - \frac{1}{2}\left(\rho^2 - r^2\right),$$
$$\frac{\partial \Phi}{\partial x} = h\left[1 + \left(\frac{r}{\rho}\right)^2\right] - x,$$
$$\frac{\partial \Phi}{\partial y} = y\left[2xh\left(\frac{r}{\rho}\right)^2 - 1\right].$$

Some physical parameters have to be imposed to evaluate the vector $\tau = [\tau_{xz}, \tau_{yz}]^T$: the radii of the beam and of the notch h and r, as well as the twist of the centroidal fiber θ and the shear modulus G. Some plausible values are considered, although they do not resemble the exact geometry of the real device. This does not affect the accuracy of the results, since this calculations are not intended to give an exact solution, for which more accurate experimental studies are needed, in the first place. However, it is thought that, by assigning proper values to the parameters, the following computations give a close enough picture of the real stress field at a section of the hinge.

From the literature [32], we set 1 µm as fixed value for r, radius of the notch. Stresses were then studied for h linearly varying from 1 µm to 20 µm with steps of 1 µm.

We then imposed:

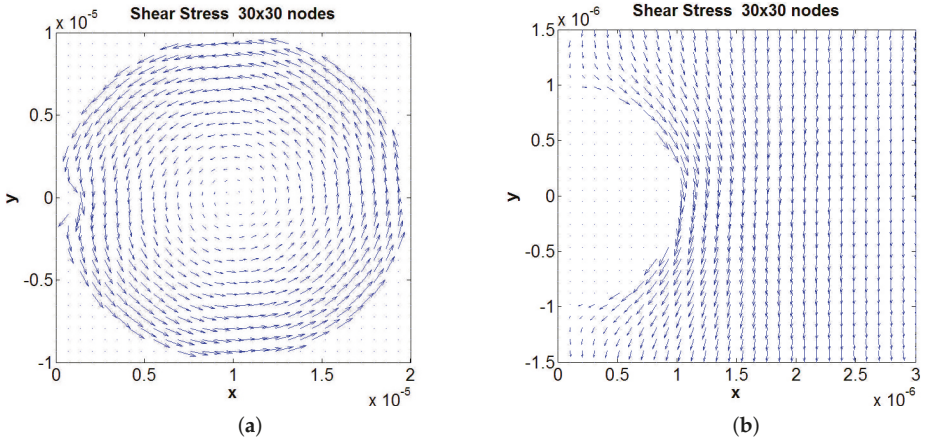
- beam radius, $h = 1 \div 20$ µm
- notch radius, $r = 1$ µm
- $G = 70$ GPa
- torsion relative angle $\omega_o = 5^\circ = 0.0873$ rad
- beam length $z = 300$ µm
- twist of the centroidal fiber $\theta = \frac{d\omega_o}{dz} = 291 \frac{\text{rad}}{\text{m}}$

The results were implemented and plotted using Matlab©. The stresses were initially computed deriving partially Φ in a 10,000 nodes sample grid in x and y over the domain D; the grid was then refined with 40,000 and, finally, 160,000 nodes.

Figure 8a,b represents the shear vector fields in the cross section and around the notch for the two cases $h = 10r$ and $h = 20r$, respectively, in different scaled grids.

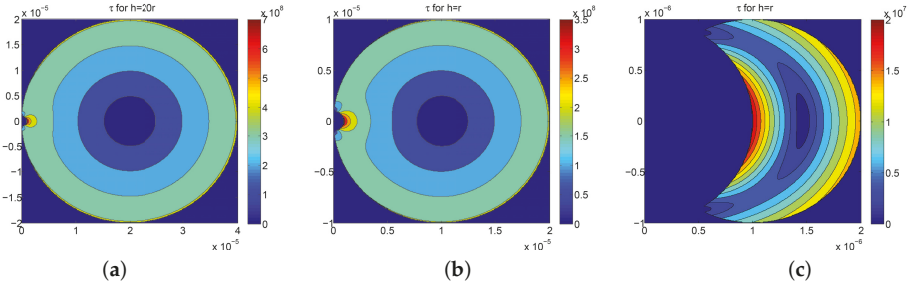
Figure 9a-c shows the shear stress field obtained by composing τ_{xz} and τ_{yz}, for different values of $\frac{h}{r}$ ratio, 20, 10 and 1, respectively.

However, for the sake of the present investigation, the area around the notch is more interesting than the bulk of the cross-section, and so a detailed computation was performed within the same coordinate system and number of nodes over a sample grid narrowed to a $3r \times 3r$ square. Results, for $h = 20r$, are shown in Figure 10a. Similar conclusions can be obtained by considering the case for which $h = r$, as depicted in Figure 10b.



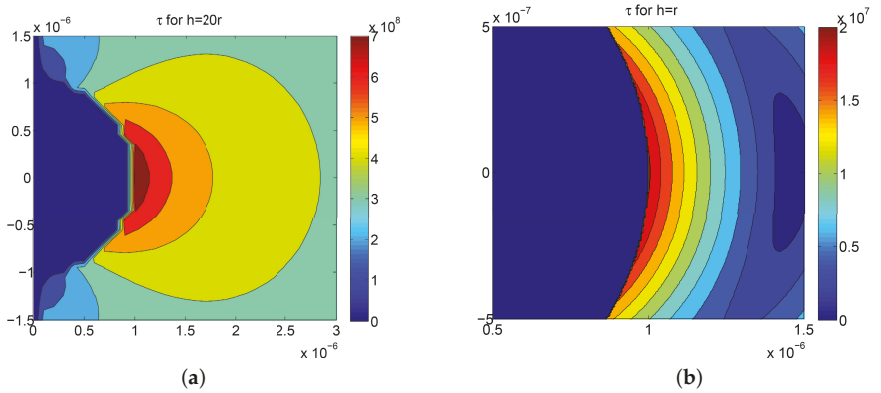
(**a**) (**b**)

Figure 8. Shear stress vector field with the analytical approach in a: (**a**) 20 µm × 20 µm grid; and (**b**) 3 µm × 3 µm grid.



(**a**) (**b**) (**c**)

Figure 9. Shear stress module (Pa) in the cross section calculated through a 400 × 400 nodes grid with the analytical approach where: $\frac{h}{r} = 20$ (**a**); $\frac{h}{r} = 10$ (**b**); and $\frac{h}{r} = 20$ (**c**).



(**a**) (**b**)

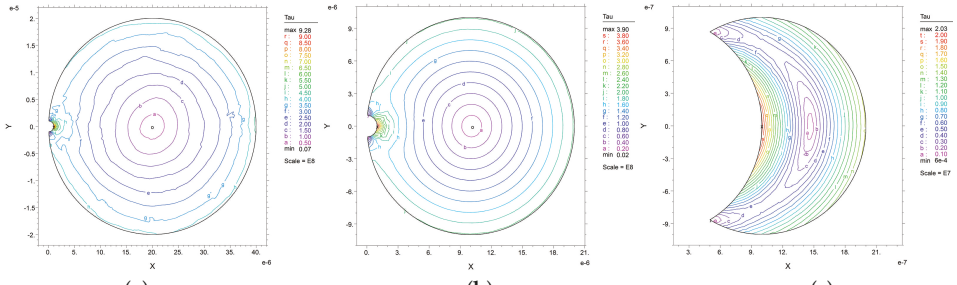
Figure 10. Shear stress module (Pa) around the notch for: $\frac{h}{r} = 20$ (**a**); and $\frac{h}{r} = 1$ (**b**).

Stress is concentrated around the circular arc midpoint, being the point where the maximum stress is attained.

7.2. Numerical Solution via Finite Difference Discretization

The computation by means of finite differences has been implemented using the free version of the software FlexPDE (*Version 6.36*).

The resulting stress field, obtained over a 587 nodes grid, is shown in Figure 11a–c, together with the plots of its two components τ_{xz} and τ_{yz}, and the stress function Φ.



(**a**) (**b**) (**c**)

Figure 11. Shear stress (10^8 Pa) map for: $h = 20r$ (**a**); $h = 10r$ (**b**); and $h = r$ (**c**).

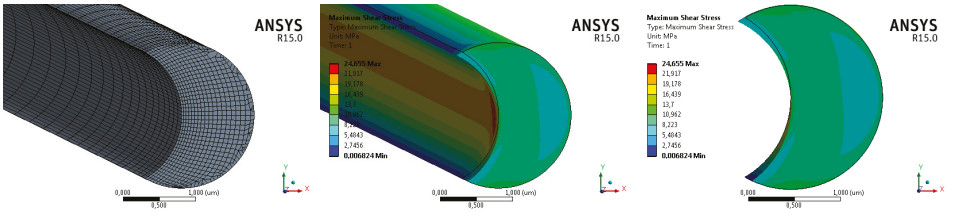
7.3. Numerical Solution via Finite Element Analysis

Finally, straight axis beams with circular section and circular notch were modeled by means of standard 3D modeling and FEA package ANSYS©.

Three different geometries were considered, thus three different solid models were built. Taking $r = 1$ µm, three cross sections with $h = 20r$, $h = 10r$ and $h = r$, respectively, were used to extrude the solid model of the beam. Using the same physical parameters in the previous methods, and imposing again 5° relative rotation among the end sections, the values of tension modules were obtained at the nodes.

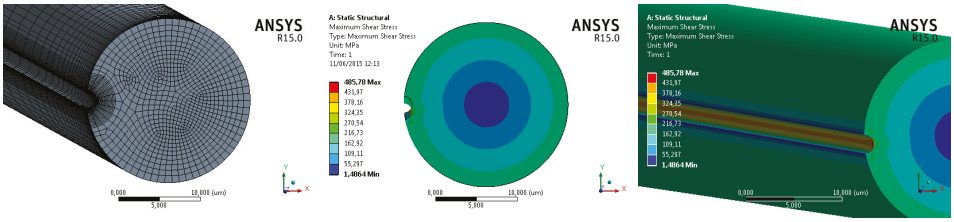
The adopted meshes, for the three h to r ratios, are represented in Figures 12a, 13a and 14a.

The maximum shear stress fields are also reported in Figure 12b,c, for the case $h = r$, while the two groups in Figures 13b,c and 14b,c concern the two cases $h = 10r$ and $h = 20r$, respectively.



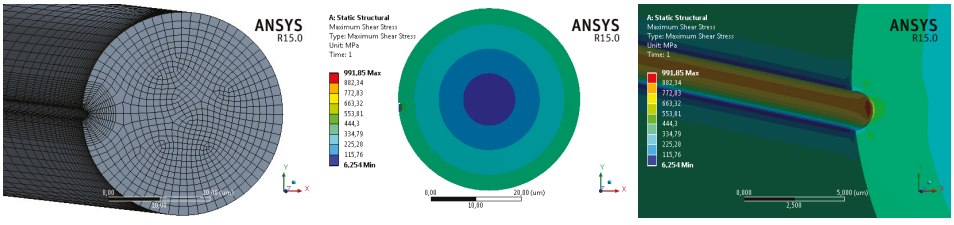
(**a**) (**b**) (**c**)

Figure 12. Summary of the results obtained by applying FEA to the case $r = h$: (**a**) mesh; (**b**) shear stress; and (**c**) shear stress in the section.



(**a**) (**b**) (**c**)

Figure 13. FEA applied to the case $h = 10r$: (**a**) mesh; and (**b**,**c**) shear stress.



(**a**) (**b**) (**c**)

Figure 14. FEA applied to the case $h = 20r$: (**a**) mesh; and (**b**,**c**) shear stress.

7.4. Discussion about the Achieved Results

The obtained results have shown that the maximum shear stress holds in the center of the notch arc. Such point is called *core point* in Table 1. On the other hand, two other points have been considered, namely, the the upper and lower edge points of the notch, which correspond to the scallops spikes. In such points, called *edge points* in Table 1, the stress (denoted with τ_{edge}) is virtually null. This is consistent with the fact that such point belongs to a *stagnation zone* of the tension field.

Table 1. Comparison of Results.

h	Point	Position		τ (MPa)			
		x	y	Analytical	Flex PDE	FEA	
						mid*	min, max*
r	Core	r	0	19.9	20.3	20.37	24.66
	Edge	$\frac{r}{2}$	$\pm\frac{\sqrt{3}}{2}r$	0	$<6.0\times10^{-3}$	6.8×10^{-3}	6.8×10^{-3}
$10r$	Core	r	0	356	390	391.93	485.78
	Edge	$\frac{r}{20}$	$\pm\frac{\sqrt{399}}{20}r$	0	<20	1.49	1.49
$20r$	Core	r	0	727	928	802.01	991.85
	Edge	$\frac{r}{40}$	$\pm\frac{\sqrt{1599}}{40}r$	0	<150	7.08	6.25

mid: referred to mid section; min, max: minimum and maximum values.

Of course, symmetry implies identical solution for the top and the bottom edge points. Shear stress has been computed in these points using three methods. The analytical solution yields null values (i.e., zero, to the best of the numerical approximation of the floating point accuracy), while the other two methods give values much smaller (practically null values) than the those calculated in the core. This difference with the analytical method is certainly due to the discretization approximation of the numerical approaches.

The proposed analytical method gave us the opportunity to calculate stress in many different points and sections. For example, τ_{edge} and $\tau_{\max}$ could be calculated as a function of the ratio ρ of the notch radius r to the circular section radius h. The results obtained have shown that the maximum stress decreases as $\rho = \frac{r}{h}$ increases. This is consistent with the physics of the phenomenon since, as h reduces in size, the discontinuity in the field becomes less severe. Considering the *edge points*, τ_{edge} also shows a similar decreasing trend as h decreases. As mentioned above, its magnitude remains significantly lower than the $\tau_{\max}$ stress achieved at the notch core. Figure 15 shows the dependency of $\tau_{\max}$ and τ_{edge} upon the $\frac{r}{h}$ ratio.

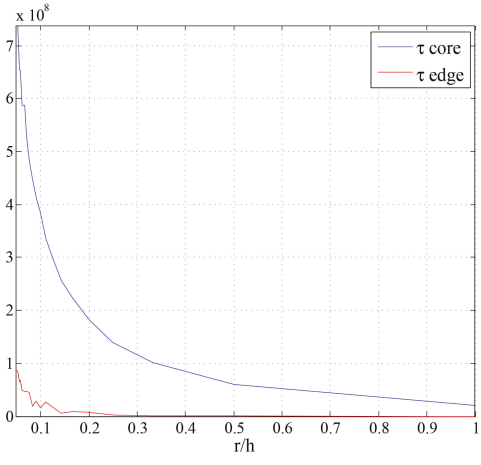


Figure 15. Stress (Pa) at the edge and at the core of the notch, as a function of $\rho = \frac{r}{h}r$.

The diagram reported in Figure 15 concerns the tension absolute values that hold in the notch core and edge as a function of ρ. However, this result would convey rather little insight if it was not reduced to a physical model *equivalent* to the scalloped notch. For this purpose, there are still two particular problems to be solved:

- the definition of an equivalence criterion for relating the adopted simplified geometry (see Figure 6) to the original geometry (see Figures 2 and 4); and
- the introduction of a reliable index of the severity of stress concentration.

7.4.1. Adopted Equivalence Criterion

One advantage of the adopted cross section is that it is very simply described by means of analytical and geometrical tools. For example, referring back again to Figure 6, it is clear that the scalloped notch is represented by the circular segment having r, d and a as radius, depth and chord, respectively. One characteristic of scalloping is the dimensionless ratio δ of the notch depth d to the chord a. The two dimensionless parameters ρ and δ can be easily related using two simple, well-known geometrical equations for the circular segment, namely,

$$\begin{aligned} d &= r - \sqrt{r^2 - \frac{a^2}{4}} \\ a &= \frac{r\sqrt{4h^2 - r^2}}{h} \end{aligned} \tag{25}$$

7.4.2. Stress Concentration Factor

The *stress concentration factor* K_t, for the considered beam, can be calculated as the ratio of the maximum stress on the notch core τ_{max} by the stress calculated on a point positioned on the other side of the beam section (with reference to Figure 6, this points is on the x axis, opposite to the origin O).

The adopted equivalence criterion and the introduction of K_t give rise to an interesting interpretation of the iterative application of the analytical method.

Figure 16 reports the values of the stress concentration factor K_t as a function of the parameter $\eta = \rho^{-1} = \frac{h}{r}$, which spans, in the real domain, from $\frac{1}{2}$ (which correspond to a beam with null cross sectional area) to ∞. There are no reasons for η to go beyond the order of magnitude of tens and so 40 has been regarded as the maximum. The trend shows a sudden increase of K_t for small increments of η. For values of η greater than about 5, K_t tends asymptotically to 2. However, according to the equivalence criterion, the scalloping geometry is better characterized by the ratio $\delta = \frac{d}{a}$.

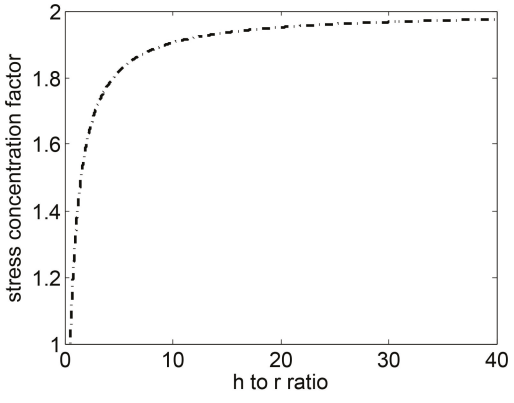


Figure 16. Stress Concentration Factor vs. $\eta = \frac{h}{r}$.

By using Equations (25), it is possible to evaluate the ratio δ of the scalloping dept d to chord a as a function of η (see Figure 17).

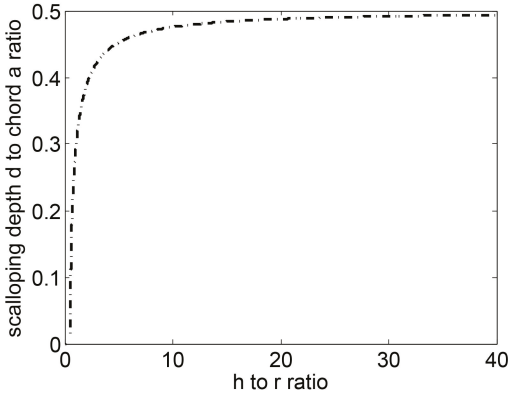


Figure 17. Ratio δ of the scalloping dept d to chord a as a function of η.

Finally, the relation between η and δ can be used to obtain the explicit dependency of K_t upon δ, as reported in Figure 18. In fact, the dependency of K_t on η and of the latter on δ, can be combined together to get, via function composition, $K_t(\delta)$.

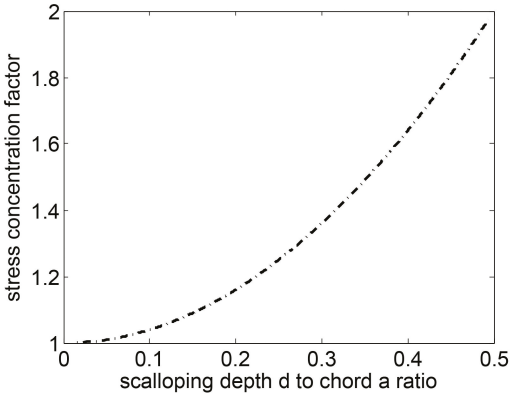


Figure 18. Stress Concentration Factor vs. δ.

When $\delta = 0$, which corresponds to a smooth wall, no stress concentration occurs ($K_t = 1$). However, for positive values of δ, the trend depicted in Figure 18 shows that the notch depth d has a great impact on stress concentration. The maximum value of K_t is 2 and it corresponds to a notch which has its center on a vertical wall (with $h >> r$).

Thanks to the above-presented analysis, it is possible to state that the stress concentration factor tends asymptotically to the theoretical value 2 as the scalloping depth increases with respect to the scalloped chord, where the limit value is already known in the literature (see, for example, Ref. [65]).

Consequently, it can be also stated, as a general principle, that, for a DRIE micromachined beam with apparent and not treated scalloping, the strength can decrease to the half value of the strength of the same beam with no notches.

8. Conclusions

The wide range of applications has made MEMS more complicated. Therefore, their mechanical structure and internal stress state also become more complex. In many of those applications, MEMS are called to interact with the environment and so they need high aspect ratios, which can be achieved by manufacturing them via DRIE process. Unfortunately, the etched profiles cannot always get rid of the scalloping problem, which induces a dangerous increase of concentrated stress in the structure. Since this stress concentration gives rise to evident loss of structural strength for silicon, it is important to take into account the effects of scalloping on stress concentration in this kind of structures.

In this paper, three different methods are proposed for stress concentration characterization of silicon DRIE micromachined structures. All of them use a straight axis beam with a circular section and a circular notch. The first approach is based on the theory of linear elasticity and evaluates the shear stresses for pure torsion (stress concentration being due to shear state only). The other two methods use numerical approach: finite difference and finite element analysis. Although the three methods are very different in nature, the obtained values are fairly comparable. Furthermore, using the analytical method makes it easier to obtain diagrams that show the stress concentration factor with respect to the notch radius. The asymptotic value of the stress concentration factor, as a function of the notch relative radius, is in accordance with the literature. This leads to the conclusion that, in DRIE manufactured notched beams, strength is virtually half of the one with no notches.

Author Contributions: Conceptualization, all authors; Methodology, S.B. and G.R.; Software, S.B. and N.P.B.; Validation, all authors; Formal Analysis, S.B., G.R. and M.V.; Investigation and resources for fabrication, A.B. and P.B.; Data curation, all authors; Writing-original draft preparation, review, editing, and visualization, M.V. and N.P.B.; and Supervision, P.B. and G.R.

Funding: This research received no external funding.

Conflicts of Interest: The authors declare no conflict of interest.

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The Effects of Structure Thickness, Air Gap Thickness and Silicon Type on the Performance of a Horizontal Electrothermal MEMS Microgripper

Marija Cauchi [1,*], Ivan Grech [2], Bertram Mallia [3], Pierluigi Mollicone [1] and Nicholas Sammut [2]

[1] Department of Mechanical Engineering, Faculty of Engineering, University of Malta, MSD 2080 Msida, Malta; pierluigi.mollicone@um.edu.mt

[2] Department of Microelectronics and Nanoelectronics, Faculty of Information and Communication Technology, University of Malta, MSD 2080 Msida, Malta; ivan.grech@um.edu.mt (I.G.); nicholas.sammut@um.edu.mt (N.S.)

[3] Department of Metallurgy and Materials Engineering, Faculty of Engineering, University of Malta, MSD 2080 Msida, Malta; bertram.mallia@um.edu.mt

* Correspondence: mcauc03@um.edu.mt; Tel.: +356-2340-2023

Received: 20 June 2018; Accepted: 11 July 2018; Published: 15 July 2018

Abstract: The ongoing development of microelectromechanical systems (MEMS) over the past decades has made possible the achievement of high-precision micromanipulation within the micromanufacturing, microassembly and biomedical fields. This paper presents different design variants of a horizontal electrothermally actuated MEMS microgripper that are developed as microsystems to micromanipulate and study the deformability properties of human red blood cells (RBCs). The presented microgripper design variants are all based on the U-shape 'hot and cold arm' actuator configuration, and are fabricated using the commercially available Multi-User MEMS Processes (MUMPs®) that are produced by MEMSCAP, Inc. (Durham, NC, USA) and that include both surface micromachined (PolyMUMPs™) and silicon-on-insulator (SOIMUMPs™) MEMS fabrication technologies. The studied microgripper design variants have the same in-plane geometry, with their main differences arising from the thickness of the fabricated structures, the consequent air gap separation between the structure and the substrate surface, as well as the intrinsic nature of the silicon material used. These factors are all inherent characteristics of the specific fabrication technologies used. PolyMUMPs™ utilises polycrystalline silicon structures that are composed of two free-standing, independently stackable structural layers, enabling the user to achieve structure thicknesses of 1.5 μm, 2 μm and 3.5 μm, respectively, whereas SOIMUMPs™ utilises a 25 μm thick single crystal silicon structure having only one free-standing structural layer. The microgripper design variants are presented and compared in this work to investigate the effect of their differences on the temperature distribution and the achieved end-effector displacement. These design variants were analytically studied, as well as numerically modelled using finite element analysis where coupled electrothermomechanical simulations were carried out in CoventorWare® (Version 10, Coventor, Inc., Cary, NC, USA). Experimental results for the microgrippers' actuation under atmospheric pressure were obtained via optical microscopy studies for the PolyMUMPs™ structures, and they were found to be conforming with the predictions of the analytical and numerical models. The focus of this work is to identify which one of the studied design variants best optimises the microgripper's electrothermomechanical performance in terms of a sufficient lateral tip displacement, minimum out-of-plane displacement at the arm tips and good heat transfer to limit the temperature at the cell gripping zone, as required for the deformability study of RBCs.

Keywords: MEMS microgrippers; micromanipulation; red blood cells; electrothermal actuation; PolyMUMPs™; SOIMUMPs™; structure thickness; air gap thickness; polysilicon; single crystal silicon

1. Introduction

The constantly growing need for high-precision micromanipulation has seen the implementation of microelectromechanical systems (MEMS) in numerous applications within the micromanufacturing and biomedical fields. Microgrippers are one such type of MEMS devices that are typically designed to safely manipulate biological cells [1–9] as well as micromechanical parts [10–12]. The successful development of a MEMS microgripper depends on taking into account several factors including the design and mechanical structure [13], the actuation mechanism [14,15], the choice of materials used [16–19], the operational requirements and limitations [20], the fabrication technology [21–26], and the environment in which the microgripper will be operated [27].

Our previous work [25] presented an analytical and a numerical model of a horizontal electrothermal microgripper design that were developed to reliably predict the temperature distribution and displacement at the arm tips when operated under atmospheric pressure. These developed models focused on an electrothermomechanical analysis which considered heat conduction to the substrate, through both the air gap and the anchor pads, as the main source of heat loss from the structure. Tip displacement results obtained from the analytical and numerical models were comparable with the results obtained from the experimental testing performed on a fabricated PolyMUMPs^{TM} structure. The main contribution of the current work consists of utilising the models established and validated in [25] to evaluate and compare the temperature and displacement behaviour of four electrothermal microgripper design variants. These microgripper design variants have the same in-plane geometry, with their main differences being the device thickness, the air gap thickness between the structure and the substrate surface, and the intrinsic nature of the silicon material, all of which are characteristics of the fabrication technologies used. In this regard, the current study investigates the feasibility of two commercially available fabrication technologies—PolyMUMPs^{TM} and SOIMUMPs^{TM}—to realise a microgripper structure with optimised electrothermomechanical performance. The electrothermomechanical performance is optimised based on the requirements to micromanipulate and study the deformability properties of human red blood cells (RBCs), and focuses on obtaining a sufficient opening stroke of the microgripper arms, minimum out-of-plane displacement at the arm tips and minimum temperature rise at the cell gripping zone.

RBCs, with a diameter of around 8 μm and an approximate thickness of 2 μm, have an average lifetime of 120 days. They are responsible for carrying oxygen, and carbon dioxide (CO_2), from the lungs to the body tissues and back, respectively. Healthy and diseased RBCs differ in their deformability properties. Healthy RBCs are capable of undergoing the necessary cellular deformation to flow even through human microcirculation vessels with small diameters, thus being able to continuously fulfill their function of supplying oxygen to the biological tissues. However, studies have shown that a number of pathological conditions such as diabetes, sickle cell anemia, malaria and other genetic disorders negatively impact the deformability properties of RBCs [28,29]. The increased cell membrane stiffness in the presence of such pathological conditions inhibits the cellular deformation required for RBCs to freely flow through the smaller blood vessels, thus preventing RBCs from performing their important function. In the case of healthy cells, the required gripping force will be of the order of a few microNewtons, with this force expected to increase to achieve the same cell deformation in diseased cells due to the increased membrane stiffness. The deformability properties of RBCs can thus be an important measure of human well-being and their measurement has in fact been the subject of various studies over the years [30,31].

This paper investigates the effects of the structure thickness, the air gap thickness, and the different sources of silicon on the performance of a horizontal electrothermal microgripper designed for the deformability study of RBCs. The implemented 'hot and cold arm' microgripper design variants and the electrothermal actuation principle are defined in Section 2. Section 3 presents and compares the commercially available surface and bulk micromachining processes whose feasibility is studied for the fabrication of the considered microgripper design variants. The developed microgripper numerical models were simulated based on the finite element analysis (FEA) modelling technique and are described in Section 4. Section 5 outlines the setup used for the experimental investigation of

the fabricated microgrippers that was carried out using optical microscopy. Section 6 presents and compares the electrical, thermal and mechanical performance of the different microgripper design variants, and finally, Section 7 highlights the concluding remarks and outcomes of this work.

2. Microgripper Design Variants and Principle of Operation

The microgripper design variants developed in this study are all derived from the U-shape 'hot and cold arm' electrothermal actuator configuration [32]. A 'hot and cold arm' actuator design is mainly composed of two parallel arms with dissimilar widths, the hot arm and the cold arm, that underlie the operating principle of the electrothermal actuator, and a flexure component that allows the actuator to bend and to achieve the required motion. Figure 1 demonstrates how two such actuators can form a complete microgripper, with the extending microgripper arms amplifying the achieved tip displacement. The 'hot and cold arm' microgripper design considered here consists of a kinematically simple structure that can be manufactured with a limited number of fabrication steps. Such a structure makes it easy to assess the feasibility of two commercially available fabrication processes as well as to study the effects of basic design parameters such as structure thickness, air gap thickness and silicon type on the performance of a microgripper for the intended application of RBC manipulation.

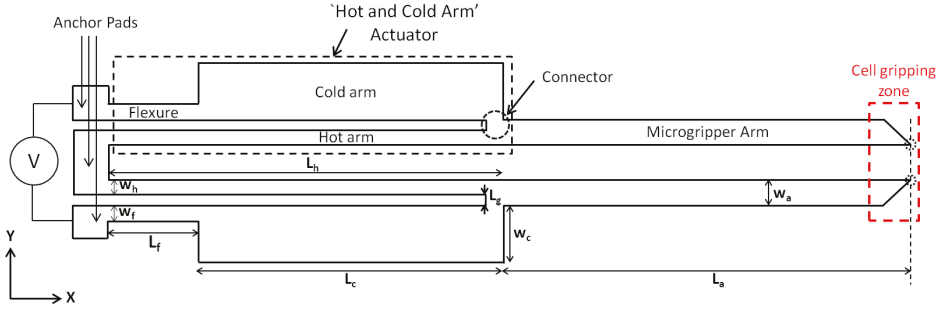


Figure 1. Schematic illustration of the electrothermal microgripper design based on the U-shape 'hot and cold arm' actuator configuration. The encircled points within the cell gripping zone indicate the arm tip points that will be later investigated for their *x*-, *y*- and *z*-displacements. The cell gripping zone is to undergo further development in future work.

Under an applied potential, the hot and cold arms experience unequal resistive heating due to their different cross-sectional areas, and the resulting asymmetric thermal expansions result in a bending moment and in angular rotation of the arms about the fixed anchor pads. This motion drives the two gripping arms to open and close as necessary, and results in non-parallel gripping of the micro-object. The relative rotation angle for the opening motion of the arms is given by the inverse tangent of the ratio of the arm tip's displacement to the combined length of the hot arm and the microgripper arm (L_h + L_a). In scenarios where parallel gripping is essential, a different kinematic design will be required [33]. There is, however, no requirement for non-parallel gripping in the case of RBC manipulation as RBCs are highly deformable and thus this non-parallel gripping is not expected to hinder their manipulation and deformability characterisation. Moreover, the same designed microgripper will be used to comparatively study the deformability properties of different RBCs, and hence the non-parallel gripping effect will not influence the results obtained.

The optimal design of flexure hinges in microgripper structures has been investigated in multiple works [34–38]. In the 'hot and cold arm' design, the ratio of the flexure length to the hot arm length is one of the design parameters that influences the maximum deflection achieved at the microgripper arm tips. This length ratio was optimised in the authors' previous work [32] and was found to have a value of 0.23. Moreover, the flexure is generally manufactured to have the same width as the hot arm,

which is in turn usually designed with the smallest dimension that the fabrication process is capable of achieving. The flexure is, however, much shorter than the hot arm, thus limiting its temperature from increasing to the same degree as that of the hot arm. These length and width criteria ascertain that the flexure is designed with the correct dimensions to effectively operate as a low-stiffness extension of the cold arm, ensuring proper elastic deflection of the actuator. As for the entire microgripper structure, the thickness of the flexure in this work is dependent on the fabrication process.

The performance of a microgripper is largely dependent on its geometry and dimensional parameters, as well as its material properties. This work focuses on four microgripper design variants that have the same dimensions in the x-y plane (i.e., the length and width parameters), but different structure and air gap thicknesses, as well as different material properties. The fixed length and width parameters of the designed microgrippers are given in Table 1.

Table 1. Fixed geometrical dimensions of the studied microgripper design variants. The dimension annotations are given in Figure 1.

Parameter	Value	Unit
Length of hot arm, L_h	200	µm
Length of cold arm, L_c	154	µm
Length of flexure, L_f	46	µm
Length of connector, L_g	3	µm
Length of gripping arm, L_a	203	µm
Width of hot arm, w_h	3	µm
Width of cold arm, w_c	14	µm
Width of flexure, w_f	3	µm
Width of gripping arm, w_a	6	µm

The differences in structure thickness, air gap thickness, and material type of the four microgripper design variants were achieved by making use of different fabrication technologies: three designs were obtained using the different structural layers offered by the PolyMUMPs™ process, and one design was obtained using the SOIMUMPs™ process. These two fabrication technologies have very different sources of silicon, mainly low pressure chemical vapour deposition (LPCVD) for polycrystalline silicon films for the PolyMUMPs™ process, and mechanically-thinned wafers of single crystal silicon (SCS) for the SOIMUMPs™ process, resulting in different material properties. SOIMUMPs™ utilises a 25 µm thick SCS structure having only one free-standing structural layer, whereas PolyMUMPs™ utilises polysilicon structures with thicknesses in the range of micrometers that are made from two free-standing, independently stackable structural layers. Although the individual sacrificial and structural layer thicknesses are determined by the PolyMUMPs™ fabrication process, three design variants with different thickness dimensions have been fabricated by using the two available structural layers individually as well as by fabricating a stack of both layers, enabling the user to achieve structure thicknesses of 1.5 µm, 2 µm and 3.5 µm with the same process. These two structural layers are both made of polysilicon but are subject to a different amount of diffusion-doping with phosphorus, leading to different values for the electrical resistivity of the two layers. For the same in-plane geometry, the different structural thicknesses result in a wide range of aspect ratio (width:thickness) values, from a value of 30 for the 1.5 µm thick PolyMUMPs™ structure down to a value of 1.8 for the 25 µm thick SOIMUMPs™ structure. Moreover, the different thicknesses of the PolyMUMPs™ structures also result in different air gap separations between the device and the substrate, while no small air gap is present in the SOIMUMPs™ structure.

3. Fabrication Processes

The PolyMUMPs™ and SOIMUMPs™ fabrication technologies are both standard, commercially available Multi-User MEMS Processes (MUMPs®) that are produced by MEMSCAP, Inc.

PolyMUMPs™ [39] is a surface micromachining process that utilises a 20 µm thick (100) SCS handle wafer (known as Substrate), a silicon nitride layer for electrical isolation (known as Nitride),

three mechanical layers of LPCVD polysilicon (known as Poly-0, Poly-1 and Poly-2), two sacrificial layers of LPCVD phosphosilicate glass (PSG) (known as Oxide-1 and Oxide-2), and a layer of gold with a thin chromium adhesion layer (known as Metal) that are deposited by electron beam for probing, bonding, or electrical routing. Once the layers are photolitographically patterned and reactive ion etched, the PolyMUMPs^{TM} process results in a suspended structure, i.e., a structure that is separated by a small air gap from the surface of the substrate. Because of this small air gap between the device and the substrate, small bumps known as dimples are introduced underneath the polysilicon beam to reduce the contact surface area in case the beam bends towards the substrate. Such dimples thus help to minimise device stiction to the substrate that is driven by surface energy. The Poly-1 and Poly-2 layers can be used individually or stacked on top of each other to create suspended movable structures that are released when the intermediate PSG layers are removed through etching with hydrofluoric (HF) solution, followed by supercritical CO_2 drying.

On the other hand, SOIMUMPs^{TM} [40] is a silicon-on-insulator (SOI) micromachining process that utilises a 400 µm thick (100) SCS handle wafer (known as Substrate), a silicon dioxide layer for electrical isolation (known as Oxide), a SCS layer that is mechanically-thinned (known as SOI), and two layers of gold with a thin chromium adhesion layer (known as Pad-Metal and Blanket-Metal) that are deposited by electron beam . The 25 µm thick SOI layer is mechanically released by jointly using an undercut etch with vapour phase HF acid to remove the Oxide layer, and deep reactive ion etching from the backside of the substrate layer (known as Trench), in this way creating an overhanging movable structure, i.e., a structure that is totally exposed to ambient with no small air gap present.

Figure 2 highlights the main dimensional differences resulting from the two fabrication processes. The main differences between structures fabricated with these two technologies are the different thickness of the released silicon layers t, the different air gap thickness t_a or lack thereof, as well as the different structural material. Table 2 summarises the microgripper design variants studied in this work.

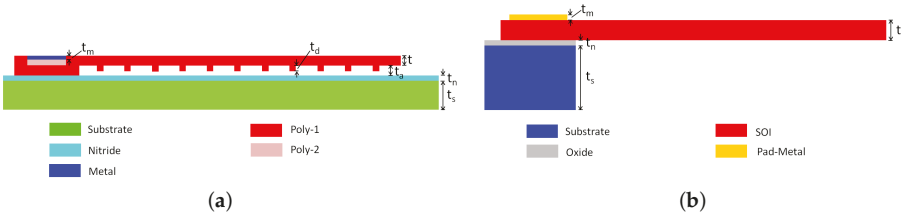


Figure 2. Cross-sectional schematic of the microgripper structure fabricated with the PolyMUMPs^{TM} surface micromachining technology (**a**) and the SOIMUMPs^{TM} bulk micromachining technology (**b**) (not-to-scale). Note that the PolyMUMPs^{TM} process results in a suspended structure (i.e., with a small air gap between the device and the substrate) while the SOIMUMPs^{TM} process results in an overhanging structure (i.e., the structure is totally exposed to ambient with no small air gap present).

Table 2. Design variants of the studied microgripper structure. The given nominal thicknesses are in µm and are annotated in Figure 2. The thickness parameters are defined by the respective fabrication process and the mask layers used.

Design Variant #	t	t_a	t_d	t_s	t_n	t_m	Process	Material	Mask Layer
1	1.5	2.75	2	20	0.6	0.5	PolyMUMPs^{TM}	Polysilicon	Poly-2
2	2	2	0.75	20	0.6	0.5	PolyMUMPs^{TM}	Polysilicon	Poly-1
3	3.5	2	0.75	20	0.6	0.5	PolyMUMPs^{TM}	Polysilicon	Poly-1 + Poly-2
4	25	-	-	400	2	0.5	SOIMUMPs^{TM}	Single Crystal Silicon	SOI

4. Numerical Models

Numerical models based on the FEA technique were developed in CoventorWare® and used to sequentially compute the steady-state electrothermal and thermomechanical performance of the designed microgrippers. The models of the microgripper design variants fabricated with the PolyMUMPs™ process (Design Variants 1-3) and with the SOIMUMPs™ process (Design Variant 4) are shown in Figures 3 and 4, respectively. A parabolic tetrahedral mesh with an element size of 1 µm was determined by a mesh refinement study to ensure an accurate and cost-effective mesh size. The numerical simulations were performed taking into account nonlinear geometry, while the influence of body forces such as gravity were not considered as they were found to be negligible at the microscale. The arm tip points indicated in Figure 1 were numerically investigated for their *x*-, *y*-, and *z*-displacements.

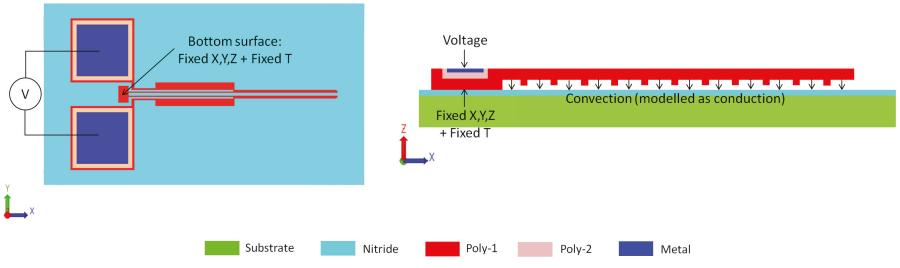


Figure 3. The suspended PolyMUMPs™ electrothermal microgripper as modelled in CoventorWare®. This model represents the mechanical structure for Design Variants 1-3, and demonstrates the boundary conditions applied to the model.

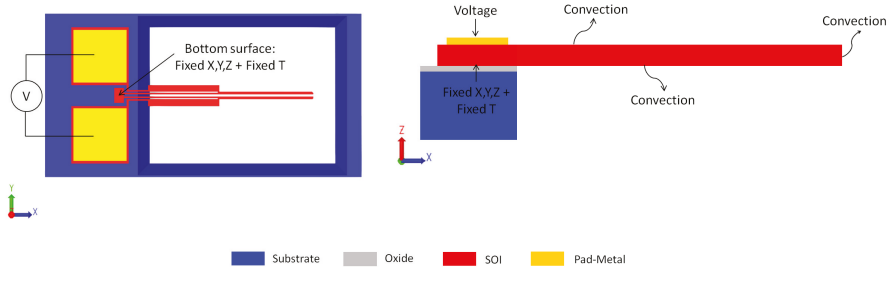


Figure 4. The overhanging SOIMUMPs™ electrothermal microgripper as modelled in CoventorWare®. This model represents the mechanical structure for Design Variant 4, and demonstrates the boundary conditions applied to the model.

The material properties used within the numerical models are given in Table 3 and Figure 5. Isotropic properties are used for polycrystalline silicon. Polysilicon films with a 1 µm thickness or larger are usually composed of relatively columnar grains with an average grain size of 300–600 nm across most of the film thickness and dominant <100> texture. This grain size is strongly influenced by deposition parameters and annealing temperatures. Such a microstructure permits large polysilicon components composed of a large aggregate of grains, such as MEMS devices that are tens to hundreds of micrometers large, to be considered as transversely isotropic and under plane strain, locally and globally [41]. In the case of SCS, while thermal properties can be assumed to be isotropic for any given plane, the elastic and mechanical properties are, however, direction dependent. This variation results from the anisotropic nature of silicon whose crystalline structure exhibits cubic symmetry. The elastic behavior of silicon MEMS structures depends on the orientation of the structure with respect to the

crystallographic axes. The (100) plane is the most common silicon wafer orientation used for the microfabrication of MEMS devices and thus the elasticity values of SCS are given in the frame of reference of a standard (100) silicon wafer in this work. Due to its cubic symmetry, it is possible to describe silicon as an orthotropic material and to give its elastic properties in terms of orthotropic material constants as shown in Table 3 [42,43].

Table 3. Material properties extracted from the CoventorWare® Materials Library for the PolyMUMPs™ [39] and SOIMUMPs™ [40] fabrication processes. Unless otherwise specified, these material properties are given at 300 K.

Property	PolyMUMPs™			SOIMUMPs™	
	Poly1	Poly2	Metal	SOI	PadMetal
Density [g/(cm)^3]	2.23	2.23	19.30	2.50	19.30
Young's modulus, E [GPa]	158	158	57	$E_x = E_y = 169$ $E_z = 130$	57
Shear modulus, G [GPa]	-	-	-	$G_{yz} = G_{zx} = 79.6$ $G_{xy} = 50.9$	-
Poisson's ratio, ν	0.22	0.22	0.35	$\nu_{yz} = 0.36$, $\nu_{zx} = 0.29$ $\nu_{xy} = 0.064$	0.35
Thermal expansion coefficient, α [µm/mK]	2.80	2.80	Refer to Figure 5	2.50	Refer to Figure 5
Specific heat capacity, c [J/kgK]	712	712	128.7	712	128.7
Thermal conductivity, k [W/mK]	32	32	297	148	297
Electrical resistivity [µ·Ω·m]	20	30	3.12×10^{-2}	500	2.86×10^{-2}

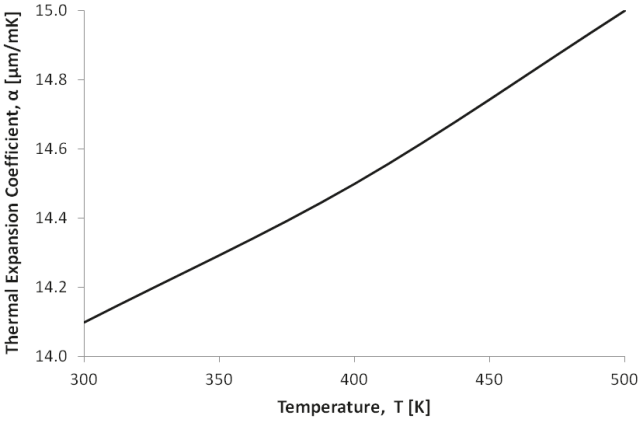


Figure 5. Variation of the coefficient of thermal expansion, α, with temperature for the Metal and PadMetal Layers in the PolyMUMPs™ [39] and SOIMUMPs™ [40] processes, respectively.

The solution of the electrothermal and thermomechanical analyses also requires a number of boundary conditions to be applied within the models to define the electrical, thermal and mechanical constraints imposed on the system. The boundary conditions implemented within these models are indicated in Figures 3 and 4, and included applying the actuation voltage to the top surface of the metallised probe pads, setting an initial temperature as a volumetric boundary condition to all the model elements, fixing the temperature of the bottom surface of the three anchored pads to room temperature throughout the analyses since these pads are anchored to the substrate, which acts as a heat sink, and restricting the motion of the bottom surface of the three anchored pads in all directions

throughout the analyses. All the above-mentioned boundary conditions were identically applied to the four microgripper design variants.

An additional boundary condition was then separately applied to the PolyMUMPs^{TM} structures (Design Variants 1-3) and the SOIMUMPs^{TM} structure (Design Variant 4) to simulate convective heat transfer. Due to the very small air gap, t_a, between the microgripper and the substrate resulting from the PolyMUMPs^{TM} fabrication technology (Figure 2a), convection from the bottom surface of the microgripper to the substrate becomes significant, and other heat losses by natural convection from the other beam surfaces can justifiably be assumed to be comparatively negligible. Moreover, the small air gap permits the convective heat transfer coefficient of the bottom surface to be modelled by conduction through the air gap to the substrate [44]. This approach was thus applied to simulate heat losses in the numerical models for Design Variants 1-3. On the other hand, due to the absence of a small air gap beneath the overhanging SOIMUMPs^{TM} structure (Figure 2b), there is no dominant convective heat loss from the bottom surface compared to the other beam surfaces and thus, in this case, a convective heat transfer coefficient was equally applied to all the beam surfaces in the numerical model for Design Variant 4 to simulate heat losses by natural convection in air. Moreover, heat losses by radiation can be reasonably ignored for thermal actuators operated at low operational power [44,45] and were thus excluded from all the models in this work.

5. Experimental Setup

The experimental testing reported in this work was carried out on the microgripper structures fabricated with PolyMUMPs^{TM} (Design Variants 1-3). Different laboratory procedures are in place to reliably test RBCs under atmospheric conditions [46], thus justifying actuation characterisation in air in this study. The probe pads of the released fabricated structures were ultrasonically wire-bonded to the contact pads of an integrated circuit (IC) package to facilitate testing. Voltage probes, each with an integrated positioning system in the x-, y- and z-directions, were then used to electrically achieve a connection between the IC package and the power supply. Each microgripper structure based on Design Variants 1-3 was actuated by applying a potential in increments of 0.5 up to 3 V on each arm, in the meantime also taking note of the current through the structure. The thermal time constant of the studied microgripper design variants is of the order of milliseconds, and thus an interval of a few seconds between each voltage step ensured that steady-state conditions were obtained before applying the new voltage.

The relative opening displacement of each microgripper arm tip was measured at atmospheric pressure on a vibration isolation Cascade Microtech Summit 11,000/12,000 B-series probe station with an embedded optical microscope-based vision system. A Mitutoyo's Metallurgical Plan Apochromatic (M Plan APO) 10× objective lens, in conjunction with the 10× magnification of the eyepiece, was used for the experimental characterisation. The features of such an objective lens include infinity corrections, bright field observation, long working distance and corrections of chromatic and spherical aberrations. A tracking software program was used to automatically measure the relative opening stroke of each arm, from which the absolute total gap opening could then be calculated by taking into account the 5 μm initial gap between the arm tips in the closed (i.e., not actuated) position. The width of the gripping arm (6 μm) was taken as a reference for calibration. The experimental setup is shown in Figure 6.

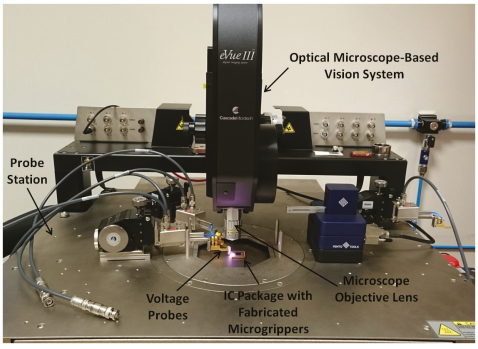


Figure 6. Experimental setup used to test the microgripper design variants fabricated with PolyMUMPs™. The microgripper structures are mounted on an integrated circuit package and actuated with voltage probes on the Cascade Microtech probe station [25].

6. Results and Discussion

The four microgripper design variants that differ in terms of the device thickness, the air gap thickness between the device and the substrate or the lack thereof, and the silicon type (polysilicon versus SCS) have been studied for their electrothermomechanical performance. This has been conducted analytically, numerically and experimentally for Design Variants 1-3, and analytically and numerically for Design Variant 4.

6.1. Thermal Analyses

The temperature distribution developed within the microgripper is one of the important factors to be considered in a microgripper design. The maximum temperature on the hot arm determines the microgripper's performance, but, at the same time, this temperature needs to be kept within the material limits to avoid the onset of material damage. Additionally, for specific applications in the biomedical field, the temperature at the microgripper arm tips needs to be kept as minimum as possible to ensure the manipulation of living cells and tissues without inducing any damage.

Thermal characterisation of the actuated microgripper structures was performed analytically and numerically in this work. Figure 7 compares the analytical results for the temperature distribution along the different actuator components for Design Variants 1-4. The temperature profiles along the hot arm $T_h(x)$, cold arm $T_c(x)$, and flexure $T_f(x)$ for these structures were calculated using Equations (1)–(3), respectively, as derived in [25]:

$$T_h(x) = T_H + C_1 e^{m_h x} + C_2 e^{-m_h x}, \tag{1}$$

$$T_c(x) = T_C + C_3 e^{m_c x} + C_4 e^{-m_c x}, \tag{2}$$

$$T_f(x) = T_F + C_5 e^{m_f x} + C_6 e^{-m_f x}, \tag{3}$$

where m_h, m_c and m_f are to be calculated for the hot arm, cold arm and flexure, respectively, using:

$$m = \sqrt{\frac{S}{k_p t R_t}}, \tag{4}$$

where S is the shape factor that accounts for the impact of the shape of the hot arm/cold arm/flexure on heat conduction to the substrate, k_p is the thermal conductivity of polysilicon, t is the device thickness and R_t is the thermal resistance between the polysilicon beam and the substrate. The constants C_1 to C_6

are calculated based upon the boundary conditions applied to the actuator structure as detailed in [25]. In the case of Design Variant 4, a very large air gap thickness of 400 μm was used in the analytical model, in this way rendering heat conduction from the bottom surface of the device through the large air gap negligible as is the case for an overhanging structure.

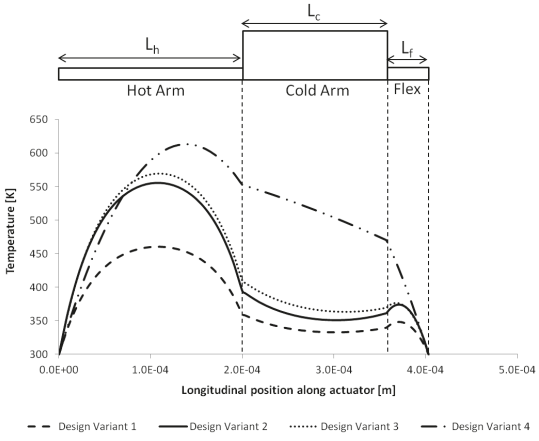


Figure 7. Comparison of the analytical results for the temperature profile along the hot arm, cold arm and flexure components of Design Variants 1-3 at 3 V, and of Design Variant 4 at 14 V. The diagram above the graph illustrates an unfolded thermal actuator without the extending microgripper arms.

The temperature distribution results obtained from the developed numerical models of the PolyMUMPs^{TM} structures (Design Variants 1-3) and the SOIMUMPs^{TM} structure (Design Variant 4) in CoventorWare® are given in Figures 8 and 9, respectively.

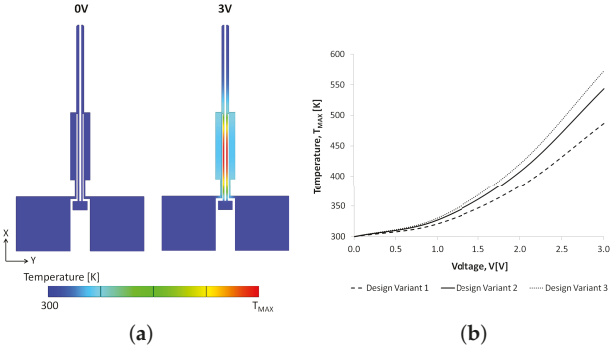


Figure 8. Simulated steady-state temperature distribution on the polysilicon microgripper structures fabricated with PolyMUMPs^{TM} (Design Variants 1-3). The simulated plots in (**a**) show the temperature distribution with and without an applied potential for Design Variants 1-3. The maximum temperature of the actuated microgripper structure is located on the hot arm and is given by T_{MAX}, which is then quantified with applied potential in (**b**) for each specific design variant.

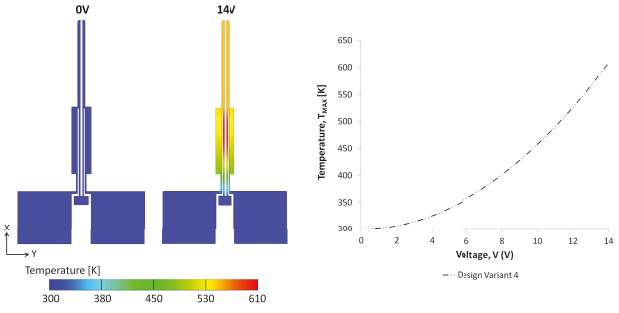


Figure 9. Simulated steady-state temperature distribution on the single crystal silicon microgripper structure fabricated with SOIMUMPs^{TM} (Design Variant 4): simulated temperature plots showing the distribution with and without an applied potential (**a**), and variation of the maximum temperature, T_{MAX}, with applied potential (**b**). The maximum temperature, T_{MAX}, of the actuated microgripper structure at 14 V is 610 K and is located on the hot arm as shown in (**a**).

It can be observed that, for each microgripper design variant, the maximum temperature is always located on the hot arm, independent of the structure thickness and the fabrication technology used (Figures 7, 8a and 9a). This is an inherent characteristic of the 'hot and cold arm' actuator design. In this design, the temperature difference between the hot and cold arms plays an important role in the achieved end-effector lateral displacement, with the latter increasing as the stated temperature difference increases for the same applied voltage. One way to maximise this temperature difference is to design the hot arm in such a way that its cross-sectional area is as small as possible considering the capability of the fabrication process. In the case of PolyMUMPs^{TM} and SOIMUMPs^{TM} fabrication technologies, the recommended minimum feature width is 3 μm [39,40], which is, in fact, the value used for the width of the hot arm for the four microgripper design variants in this study (Table 1).

When considering the microgrippers fabricated with the PolyMUMPs^{TM} technology (Design Variants 1-3), it can be observed that the most distinguished difference is between the temperature distribution of Design Variant 1, and that of Design Variants 2 and 3. The structure thickness, the air gap thickness and the intrinsic nature of the materials used (mainly in terms of the electrical resistivity property) all play a major role in defining the obtained temperature distribution on the microgripper structure. The maximum temperature developed on the hot arm increases with a thicker structure, a larger air gap thickness due to less heat lost by conduction through the air gap, and a smaller value for the electrical resistivity material property. When compared to Design Variants 2 and 3, Design Variant 1 has the smallest structure thickness of 1.5 μm, the largest air gap thickness of 2.75 μm and the highest value for electrical resistivity (Table 3). The effects of the higher electrical resistivity value and the smaller structure thickness outweigh the effect of the larger air gap thickness, resulting in an overall reduction in the maximum temperature on the hot arm. The temperature distribution for Design Variants 2 and 3 is similar. These two microgripper design variants have the same air gap thickness (2 μm), which affects the amount of heat lost by conduction to the substrate. The different thicknesses (2 μm for Design Variant 2 and 3.5 μm for Design Variant 3) and electrical resistivities (20 μΩm for Design Variant 2 and 23.33 μΩm for Design Variant 3) of the two structures almost balance out each other, resulting in a similar temperature distribution.

On comparing the temperature distribution of a surface micromachined structure (Design Variants 1-3) as given in Figures 7 and 8, to that of a bulk micromachined structure (Design Variant 4) as given in Figures 7 and 9, it can be observed that, despite the larger structure thickness and lack of a small air gap within the SOIMUMPs^{TM} structure, the larger electrical resistivity value of SCS when compared to that of polysilicon (Table 3) results in greater power required to achieve a quantitatively

similar temperature distribution to that obtained for the PolyMUMPs™ structures. It can be also observed from Figure 7 that the temperatures on the cold arm and the flexure are relatively high for Design Variant 4 when compared to those for Design Variants 1-3. This results in Design Variant 4 having the smallest temperature difference between the hot and cold arms, and, as stated previously, the larger this temperature difference, the larger will be the achieved end-effector lateral displacement. This small temperature difference will in fact cause the tips of Design Variant 4 to undergo the smallest opening stroke as will be shown in Section 6.2. Another interesting observation is that while the arm tips of the actuated PolyMUMPs™ structures remain at room temperature (Figure 8a), this is not the case for the SOIMUMPs™ structure. It is shown in Figure 9a that the arm tips of the actuated SOIMUMPs™ microgripper undergo a rise in temperature due to the higher thermal conductivity of SCS when compared to that of polysilicon (Table 3). The lower thermal conductivity value of polycrystalline silicon is a result of the grain boundary scattering of phonons that governs the thermal resistance [47]. It must be ensured that the temperature rise at the arm tips is limited as much as possible to prevent damage to the studied RBCs.

6.2. Structural Analyses

A horizontal MEMS microgripper should always be designed such that it maximises the in-plane displacement and minimises the out-of-plane deflection at the microgripper am tips. The steady-state lateral displacement at the gripping arm tip, u_{tip}, was calculated as a function of applied potential using Equation (5) as derived in [25]:

$$u_{tip} = \frac{{L_h}^2}{6EI_h}\left(X_1 L_h - 3X_3\right) + \frac{L_a L_h}{2EI_h}\left(L_h X_1 - 2X_3\right), \tag{5}$$

where L_h is the length of the hot arm, E is the Young's modulus of polysilicon, I_h is the moment of inertia for the hot arm, and X_1, X_2 and X_3 are the chosen redundant loads acting on the actuator structure. The force method [48] was applied to analyse the bending moment of the actuator due to these three redundant loads, which were calculated by solving the following set of simultaneous equations as detailed in [25]:

$$\begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} = \begin{bmatrix} 0 \\ \Delta L_h - \Delta L_c - \Delta L_f \\ 0 \end{bmatrix}, \tag{6}$$

where the terms f_{ij} represent flexibility coefficients that define the deflection at i due to a unit load at j. The flexibility coefficients are determined using Equation (7) where m_i and m_j represent the bending moments due to the respective three unit redundants X_1, X_2 and X_3:

$$f_{ij} = \int_L \frac{m_i m_j}{EI} dx. \tag{7}$$

The relative opening stroke of each arm tip, calculated using Equation (5), is compared for Design Variants 1-4 in Figure 10. The steady-state in-plane and out-of-plane displacements are also obtained numerically with CoventorWare® for Design Variants 1-4 (Figures 11–13), and good agreement is observed for the obtained in-plane displacement between the analytical and numerical results. The current and power dissipated across each microgripper arm of Design Variants 1-4 with applied voltage are given in Figure 14.

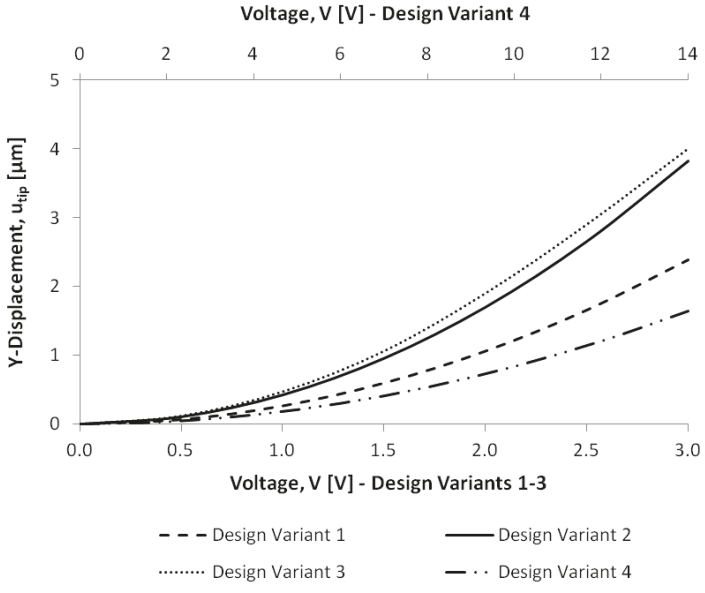


Figure 10. Analytical results for the variation of the relative y-displacement of the arm tip, u_{tip}, with applied voltage for Design Variants 1-4.

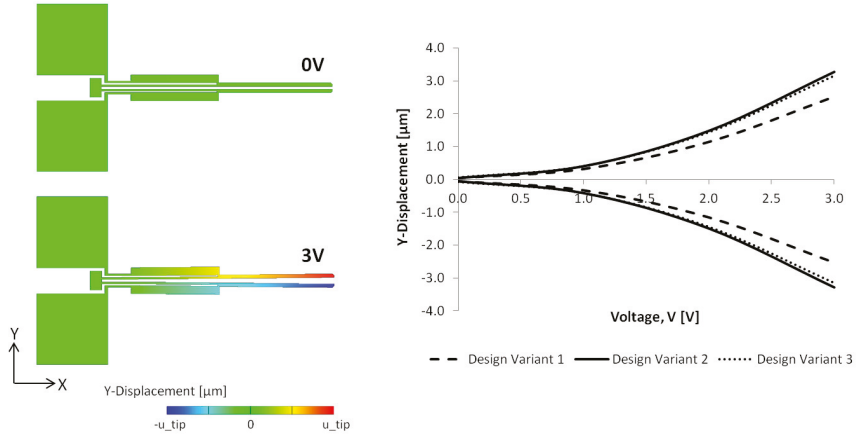


Figure 11. Simulated steady-state lateral displacement distribution on the polysilicon microgripper structures fabricated with PolyMUMPs™ (Design Variants 1-3). The simulated plots in (**a**) show the y-displacement distribution with and without an applied potential for Design Variants 1-3. The absolute gap opening in the closed position (i.e., when not actuated), u_o, is 5 µm with this absolute gap increasing to $u_o + 2u_{tip}$ when actuated. The relative opening stroke of each microgripper arm tip is then quantified with applied potential in (**b**) for each specific design variant.

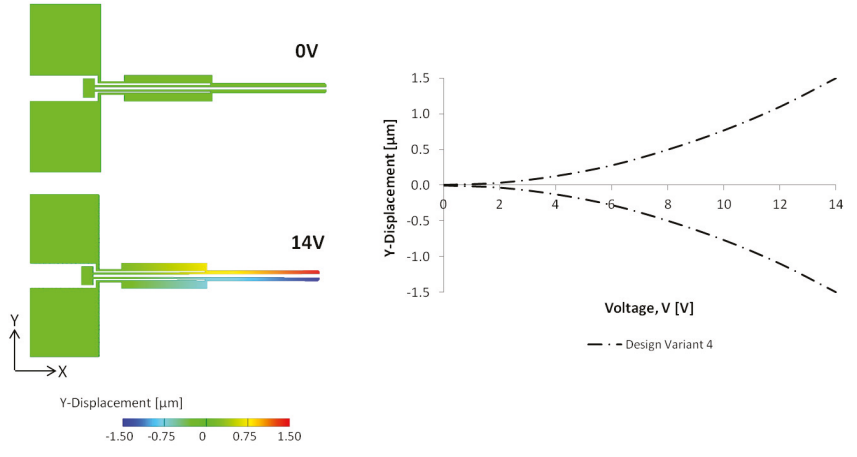


Figure 12. Simulated steady-state lateral displacement distribution on the single crystal silicon microgripper structure fabricated with SOIMUMPs^{TM} (Design 4): simulated *y*-displacement plots showing the distribution with and without an applied potential (**a**); and variation of the relative opening stroke of each microgripper arm tip with applied voltage (**b**). The absolute gap opening in the closed position is 5 µm with this absolute gap increasing to 8 µm at 14 V.

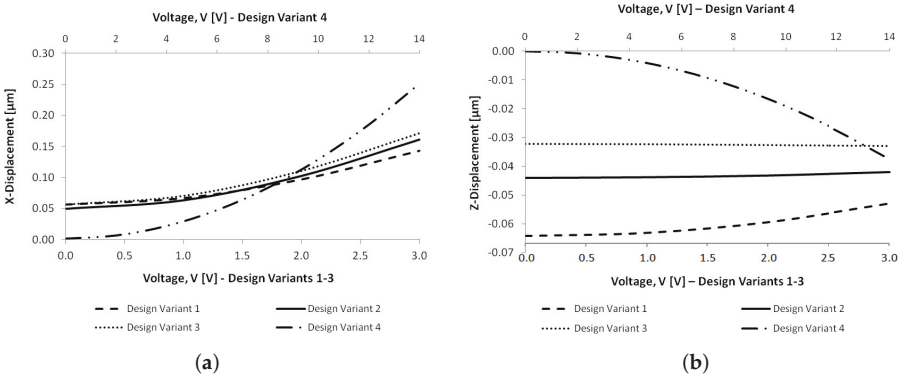
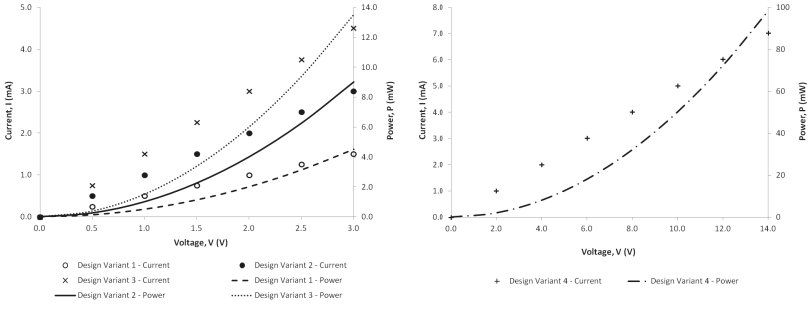


Figure 13. Simulated steady-state displacement profiles of Design Variants 1-4 with applied potential: *x*-displacement (forward displacement) of the microgripper arm tips (**a**); and *z*-displacement (out-of-plane displacement) of the microgripper arm tips (**b**).

On comparing the relative arm lateral displacements of Design Variants 1-4 (Figures 10–12), it is evident that Design Variant 4 undergoes the smallest opening stroke when actuated. In the case of Design Variant 4, the total gap opening is analytically and numerically predicted to be around 8 µm at an applied potential of 14 V, which is just in line with the design requirement of gripping a RBC. On the other hand, Design Variants 1-3 exceed the basic requirement of 8 µm for the absolute gap opening between the arm tips. Despite this, it can be observed that SOIMUMPs^{TM} structures require much greater power than PolyMUMPs^{TM} structures (Figure 14) to achieve the intended motion. Another important observation is the difference in out-of-plane displacement (*z*-displacement) at 0 V for Design Variants 1-4 (Figure 13b). This difference can be attributed to the inherent greater thickness

and lower aspect ratio (width:thickness) fabrication of SOIMUMPs^{TM} structures when compared to PolyMUMPs^{TM} structures. The aspect ratio for PolyMUMPs^{TM} structures depends on the layers used, but it is always much larger than the aspect ratio for SOIMUMPs^{TM} structures. The greater moment of inertia of structures with low aspect ratio, such as those fabricated with SOIMUMPs^{TM}, minimises any unwanted out-of-plane displacement of the actuator arms as well as offers better resistance to the effects of in-plane residual stress, making such structures more suitable for in-plane directed thermal actuators [49]. It can also be observed from Figure 13b that Design Variant 3, which is composed of the stacked Poly-1 and Poly-2 layers, has the least out-of-plane displacement when compared to Design Variants 1 and 2, which confirms that lower aspect ratio structures minimise out-of-plane bending occurring as a result of residual stresses inherent in the fabrication process. Minimising the out-of-plane path deviation of the microgripper arms is an important requirement for the successful gripping of RBCs to ensure that the arm tips remain on the same plane as the RBC so that the latter can be successfully gripped during actuation.



(**a**) (**b**)

Figure 14. Numerical variation of the current and power dissipated with voltage across each arm of the microgripper structure for Design Variants 1-3 (**a**), and for Design Variant 4 (**b**).

Figure 15 shows the investigation of the developed maximum stresses within the 'hot and cold arm' microgripper structure. It can be observed that the maximum stresses are mainly located where the arms join the anchor pads, within the flexure component and at the connectors joining the hot and cold arms. The maximum Von Mises stress for Design Variants 1-4 ranges between 40 MPa and 60 MPa at full actuation voltage (3 V for Design Variants 1-3, 14 V for Design Variant 4). The ultimate tensile strength for polysilicon and SCS has been reported in literature as 2.9 GPa [50] and 2 GPa [51], respectively. This indicates that the simulated maximum stress values are by far within the limits of the allowable stress, concluding that maximum stresses and strains during the opening motion are not critical.

The analytical and numerical models of the PolyMUMPs^{TM} structures (Design Variants 1-3) were validated through optical microscopy studies carried out under atmospheric pressure using the setup described in Section 5. Figure 16 shows the optical microscope images of Design 2 in the closed and actuated positions. The absolute gap opening between the arm tips when the microgripper is not actuated is 5 μm. The simulation and experimental measurement results of the absolute gap opening between the microgripper arm tips are compared for Design Variants 1-4 in Figure 17. Figure 17 demonstrates that the mean absolute gap openings obtained experimentally at 3 V are 9.5 μm for Design 1, 12.5 μm for Design 2 and 12 μm for Design 3, and that the results predicted by the numerical models are in good agreement with the experimental results for Design Variants 1-3. The measured gap opening values confirm that Design Variants 1-3 are all suitable for the desired function of securing a RBC with an approximate 8 μm diameter.

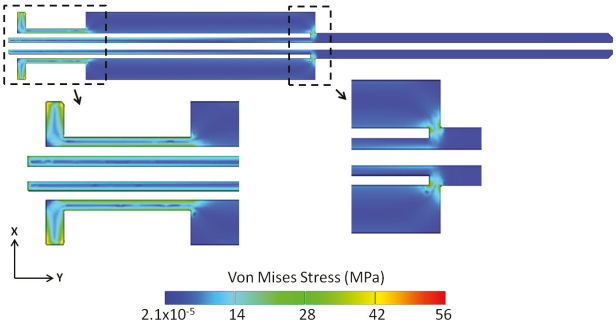


Figure 15. Von Mises stresses developed within the microgripper structure for Design Variant 1 at 3 V with inset images showing detailed views of the stresses where the arms join the anchor pads, within the flexure component and at the connectors joining the hot and cold arms.

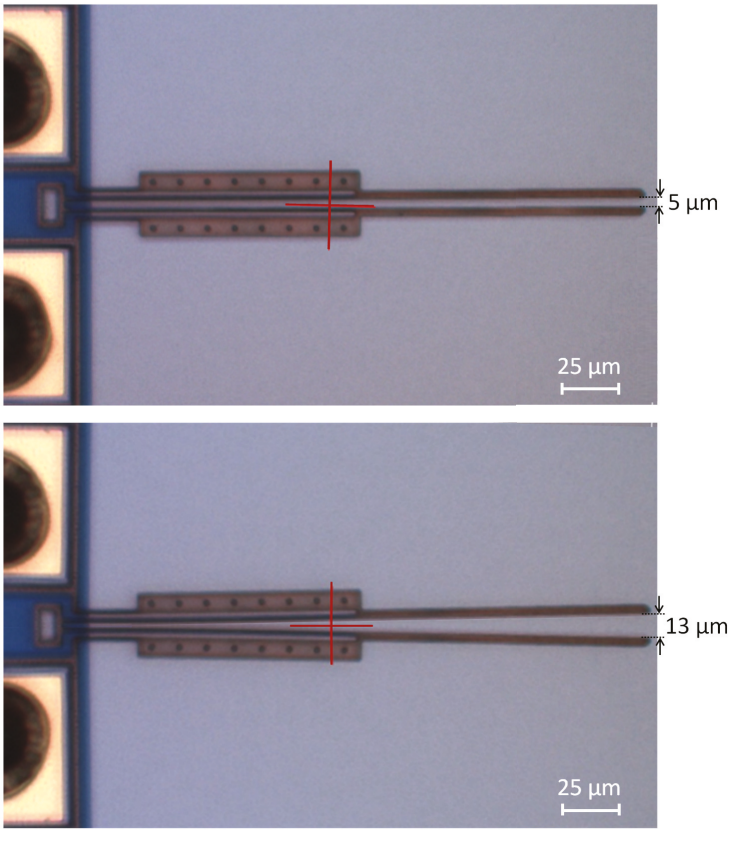


Figure 16. Images based on optical microscopy of one of the fabricated microgripper structures (Design 2) when not actuated (**top**), and when each arm is electrothermally actuated with 3 V (**bottom**). The pixel resolution of the vision system photographs is 1023 pixels (width) × 767 pixels (height), and the marked values give the absolute gap openings between the microgripper arm tips.

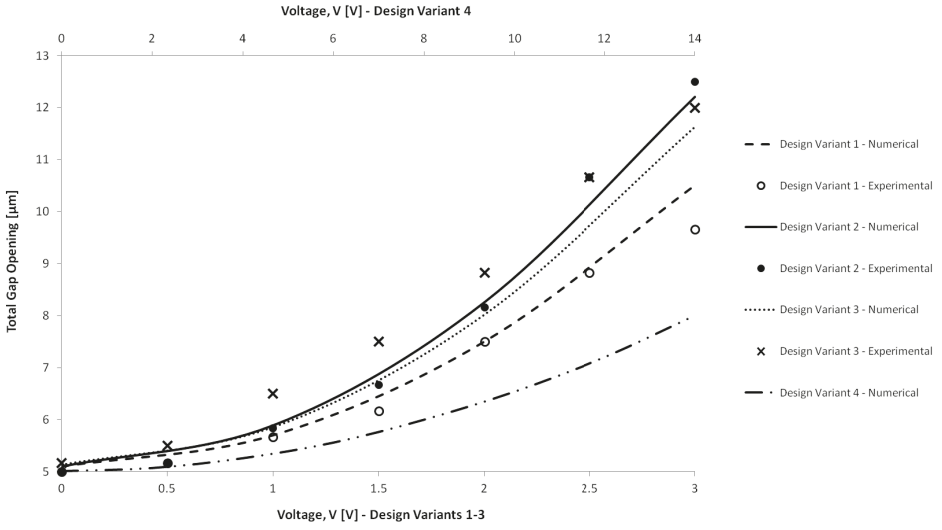


Figure 17. Comparison of the experimental and numerical results for the absolute gap opening between the arm tips of the studied microgripper design variants with applied voltage. Experimental (mean values) and numerical results are presented for Design Variants 1-3 while numerical results are shown for Design Variant 4. All design variants start from an absolute gap opening of 5 μm, which is the gap opening in the closed position.

7. Conclusions

This paper has presented four design variants of a horizontal electrothermally actuated MEMS microgripper that were developed for the micromanipulation and deformability study of RBCs using two commercially available fabrication technologies. Three microgripper design variants were achieved using the different stackable structural layers offered by the PolyMUMPs^{TM} process, resulting in polysilicon structure thicknesses of 1.5 μm, 2 μm and 3.5 μm. The other microgripper design variant was achieved using the SOIMUMPs^{TM} process resulting in a SCS structure with a 25 μm thickness. The studied microgripper design variants have the same in-plane geometry and their main differences are due to the mechanical structure thickness, the air gap thickness or lack thereof, and the silicon type (polysilicon versus SCS), all of which are a result of the inherent characteristics of the specific fabrication technology used.

The electrothermomechanical performance, specifically the power dissipation, the temperature distribution, the lateral opening stroke and the out-of-plane deflection at the microgripper arm tips, were examined and compared for the microgripper design variants. Analytical calculations and FEA simulations with the commercial software package CoventorWare® were performed on the four microgripper design variants to investigate the steady-state response of the microgripper structures under electrothermal actuation. It could be observed that greater power is required for the SOIMUMPs^{TM} structure to achieve a quantitatively similar temperature distribution and tip displacement to that obtained from the PolyMUMPs^{TM} structures. Another interesting observation was that the microgripper arm tips of the SOIMUMPs^{TM} structure, unlike PolyMUMPs^{TM} structures, undergo a rise in temperature, making such a design less suitable for the safe manipulation of biological cells. However, SOIMUMPs^{TM} structures have lower aspect ratios when compared to PolyMUMPs^{TM} structures. This means that the greater moment of inertia of SOIMUMPs^{TM} structures results in substantial out-of-plane stiffness, thus minimising any undesired out-of-plane path deviation and ensuring a more stable in-plane motion. The maximum stresses developed within the microgripper

structure were also investigated for all the design variants and they were found to be well within the allowable material limits, confirming that the microgripper design variants are able to perform a stable opening and closing motion.

The three microgripper design variants based on the PolyMUMPs^{TM} process were fabricated, and were subject to optical microscopy studies under atmospheric pressure. The tip displacement results from the actuation testing were found to comply with the analytical and numerical predictions, and all the design variants were also found to be in line with the design requirement to secure and characterise a RBC. However, it can be concluded that Design 3, which consists of two polysilicon stackable layers, resulting in a 3.5 µm thick PolyMUMPs^{TM} structure, best fulfills the desired function. This design experimentally achieved a sufficient mean gap opening of 12 µm at 3 V, while, at the same time, it was numerically found to limit the out-of-plane motion at the arm tips and to ensure room temperature conditions at the cell gripping zone.

Author Contributions: Conceptualisation, M.C., I.G., B.M., P.M. and N.S.; Formal analysis, M.C.; Funding acquisition, M.C., I.G., B.M., P.M. and N.S.; Investigation, M.C.; Methodology, M.C.; Project administration, M.C., I.G., B.M., P.M. and N.S.; Resources, M.C.; Supervision, I.G., B.M., P.M. and N.S.; Validation, M.C.; Visualisation, M.C.; Writing—Original draft, M.C.; Writing—Review and editing, M.C., I.G., B.M., P.M. and N.S.

Funding: The research work disclosed in this publication is funded by the Reach High Scholars Programme— Post-Doctoral Grants. The grant is part-financed by the European Union, Operational Programme II—Cohesion Policy 2014–2020 Investing in human capital to create more opportunities and promote the wellbeing of society— European Social Fund (ESF): grant number 254/15/05.

Acknowledgments: The technical help of Barnaby Portelli in the experimental activities performed at the Microelectronics Lab is acknowledged.

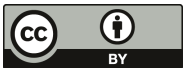
Conflicts of Interest: The authors declare no conflict of interest.

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Polymer Microgripper with Autofocusing and Visual Tracking Operations to Grip Particle Moving in Liquid

Ren-Jung Chang * and Yu-Cheng Chien

Department of Mechanical Engineering, National Cheng Kung University, Tainan 70101, Taiwan; kissme221017@hotmail.com

* Correspondence: rjchang@mail.ncku.edu.tw; Tel.: +886-6-275-7575-2651

Received: 21 May 2018; Accepted: 7 June 2018; Published: 11 June 2018

Abstract: A visual-servo automatic micromanipulating system was developed and tested for gripping the moving microparticle suspended in liquid well. An innovative design of microgripper integrated with flexible arms was utilized to constrain particles in a moving work space. A novel focus function by non-normalized wavelet entropy was proposed and utilized to estimate the depth for the alignment of microgripper tips and moving particle in the same focus plane. An enhanced tracking algorithm, which is based on Polar Coordinate System Similarity, incorporated with template matching, edge detection method, and circular Hough Transform, was implemented. Experimental tests of the manipulation processes from moving gripper to tracking, gripping, transporting, and releasing 30–50 μm Polystyrene particle in 25 °C water were carried out.

Keywords: microgripper; visual servo; moving particle; autofocus; tracking; gripping

1. Introduction

The manipulation of micro objects in liquid environment has become a great challenge in biotechnical engineering [1]. Micromanipulation of artificial particle or biological cell is a crucial technology for acquiring the geometrical, mechanical, electrical and/or chemical properties of micro objects. In addition, micromanipulation technique can be employed for analyzing the interactions between artificial particle and biological object. Micromanipulation in general requires accurate and precise position and/or force operations in holding, picking, placing, pulling, pushing, cutting, and indenting biotechnical micro objects. Regarding biotechnical manipulation, various approaches have been implemented: optic and electric micromanipulation, magnetic micromanipulation, microelectromechanical systems (MEMS) and mechanical micromanipulation [2]. Early research on utilizing micro end effector for biotechnical manipulation was contributed by Carrozza et al. [3], Ok et al. [4], and Arai et al. [5]. For manipulating moving bioparticle or cell, a non-damage holding technique is generally required before proceeding with further micro operation. In the conventional micromanipulation, a pipette is utilized carefully to hold the micro objects in liquid. In a more recent approach, acoustic pressure field is proposed to preposition micro particles in liquid prior to manipulation by utilizing MEMS gripper [6]. To work with biological samples, such as living cells, it must be possible to immerse the end effector tips into liquids. The micromanipulation technology with device dimensions and material properties compatible to biological cell finds important applications in the biomedical manipulation [7,8]. In considering the different requirements of manipulating micro particles in liquid environment, several mechanical microgrippers have been developed for these applications [1,3,6–18]. Regarding the manipulation of micro particles in liquid environment, the research on gripping stationary artificial particle [1,3,6,7,10–12,15] and living cell [3,6,9,13,14,16] has been reported. For the in vivo operations at smaller size scales, untethered

microgrippers that can be navigated to grasp living cell or tissue have been reviewed recently [19,20]. When the micro living cell is moving in liquid, the manipulation technique to hold the cell without causing damage needs further development.

Microscopic autofocus operation is an important technique utilized in the detection, tracking, and grasping of micro objects [21–24]. In gripping objects, it is actually one of the most essential techniques to ensure the successful operation. In the literature, the performance of focus functions has been widely investigated and compared [25–31]. Based on the comparison results of focusing different types of images, in general, recommended effective focus functions include energy Laplace [25], normalized variance [30], wavelet-based measure [31], and autocorrelation for fluorescence microscopy [29]. Regarding the technique for tracking particle, it can be traced back to the Particle Tracking Velocimetry (PTV). In the literature, there are several tracking algorithms for PTV ([32], and references therein). In implementing the digital PTV, temporal tracking scheme of matching the same particles in the consecutive time step is the core technique of each PTV algorithm. A tracking algorithm, which is insensitive to experimental parameters and applied to flows subjected to drift and diffusion, is considered in this paper. Although the performance of focusing functions has been widely tested in the aforementioned literature, it is noted that the focusing operation has not been tested by integrating them with visual tracking and grasping operation. Particularly, the automatic alignment of microgripper tips with moving micro particle in the same focus plane is the most essential test to ensure successful gripping operation.

In this paper, innovative design of microgripper, novel focus function, and enhanced tracking algorithm are proposed for gripping micro particle moving in liquid. After the Introduction, Section 2 describes the system design and installation of the micro manipulation system. An innovative design of microgripper integrated with flexible arms is utilized to constrain particles in a moving work space. Section 3 is dedicated to the innovative autofocus function which is utilized to estimate the depth for the alignment of microgripper tips and moving particle in the same focus plane. Section 4 gives an enhanced two-frame scheme for tracking particle under drift and diffusion flow. Section 5 provides experimental test and record of gripping microparticle. Operational processes from moving gripper to tracking, gripping, transporting, and releasing micro particle were carried out in experimental tests. In Section 6, we summarize the present innovative approach of utilizing microgripper to grip moving particle.

2. System Design and Installation

A micromanipulation system installed to realize grasping micro particle with autofocusing and automatic tracking operations is shown in Figure 1. The micromanipulation system consists of Micro Gripper System, Moving Manipulator System, Object Stage, and Autofocusing Stage. The detail design of the four subsystems is described in the following sections.

2.1. Micro Gripper System and Moving Manipulator System

Micro Gripper System was installed on the Moving Manipulator System. The Moving Manipulator System, MMS-77 from Shimadzu (Kyoto, Japan), is an *X*-*Y*-*Z* stage with 0.1 μm resolution, 30 mm stroke for *X* and *Y* axes, and 15 mm stroke for *Z* axis. The maximum moving speed is 3 mm/s. The Micro Gripper System consists of Gripper and Actuator Mechanism as well as Gripper Controller. The Gripper and Actuator Mechanism include micro gripper and actuating mechanism, as shown in Figure 2. The micro gripper is fabricated by utilizing Excimer Laser, Exitech 2000 (Oxford, UK), with mask projection through 10× size reduction by an optical lens. The micro gripper is driven through a connecting rod by piezoelectric actuator, P-820.10. The design with pushing rod is to minimize the geometrical interference between the piezoelectric actuator and liquid well in gripping operation. By employing the present design, the Micro Gripper System is able to grip a micro particle with diameter around 20–90 μm by the input of 0–100 V from the Gripper Controller, PI (Physik Instrumente) E-503.00 (Munich, Germany).

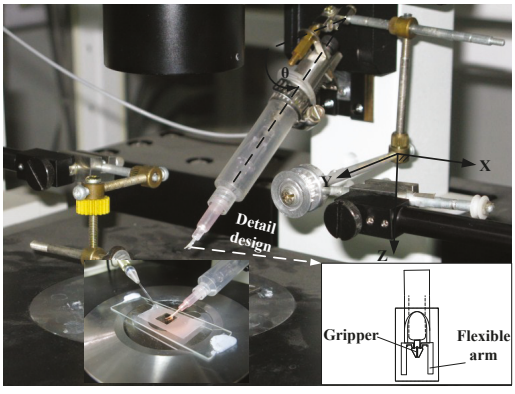


Figure 1. Micromanipulation system (overlapped by a photo of microgripper with bending flexible arms in liquid well).

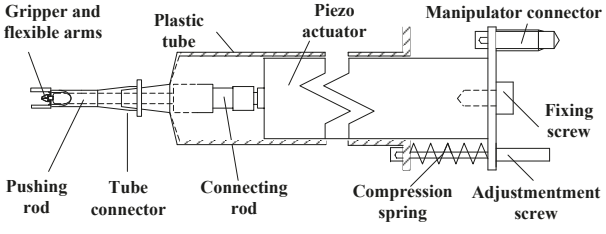


Figure 2. Detail components and assembly of Gripper and Actuator Mechanism.

The innovative micro gripper is made of PU (Polyurethane) film which is 200 μm in thickness. The Young's modulus of PU is 76.4 MPa. The present gripper mechanism is similar to that of the SMA (Shape Memory Alloy) drive PU microgripper employed for gripping micro particle [33]. The mechanism of micro gripper is symmetric with respect to the gripper axis. The kinematic motion of one gripper tip can be modeled through a four-bar orthogonal double slider linkage with one slider along gripper axis as input. Regarding the gripping force, it depends on the geometrical and material property of micro object, gripper–object contact condition, as well as the holding and gripping displacement. When gripping a soft object with Young's modulus less than that of the PU, the gripping force can be analyzed and calibrated in advance through controlling the input displacement. In the present innovative design, as shown in the bottom right corner of Figure 1, the design is to incorporate a pair of flexible arms which are fabricated and integrated on the outer right and left sides of gripper frame. The flexible arms are designed to provide important functions: visual contact probe, constraint of particle movement, and reference for focus plane. When the manipulator moves downward in Z direction, the flexible arms will first approach to touch the inner bottom of liquid container. Then, the continue motion of Moving Manipulator System will cause bending of the flexible arms. The bending flexible arms form a rectangular notch to constrain the fluid flow and trap moving particles. In addition, the reflected edge of bending flexible arms can be monitored through the vertical visual CCD (Charge Coupled Device) system to provide a reference focus plane for positioning the gripper tips and avoiding collision with bottom surface.

2.2. Object Platform

Object Platform is a platform to support particles and fluid container. The platform was installed on another MMS-77 from Shimadzu (Kyoto, Japan). The specifications of MMS-77 to support container are the same as those of MMS-77 employed for installing Micro Gripper System.

2.3. Autofocusing Stage

Autofocusing Stage includes illumination system, CCD visual system, *Z*-axis servo system, and autofocusing controller. The autofocusing system is implemented through Olympus IX71 (Tokyo, Japan) inverted microscope. With 4× object lens, the view range is 1592 × 1194 × 25 μm^3. The stage is driven by stepping motor with 1.8 degree/step through a gear train of 4.54 ratio. The moving resolution of focusing stage in *Z* axis by stepping motor is 0.28 μm/degree. The detail components of the image-based Autofocusing Stage are shown in the block diagram of Figure 3. A functional block of the Procedure of autofocusing operation is implemented by PC (Personal computer). The image signal of moving *Z* position is sensed by CCD through microscope and image capture card. The focusing control through PC can be operated in open-loop or closed-loop mode. The closed-loop control is to realize the autofocusing operation with feedback image signal of moving *Z* position.

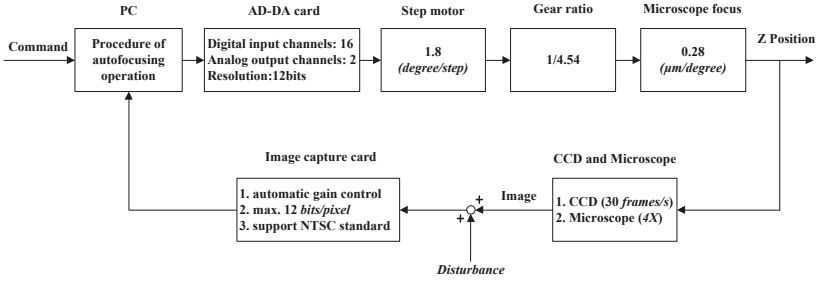


Figure 3. Block diagram of focusing control in open or closed loop through PC.

The synergetic operation of the four subsystems for automatic operation of focusing, tracking and grasping micro particle is implemented by utilizing C/C++ on a PC with Intel® Core™ 2 CPU. The Clock rate of CPU is 2.13 GHz. The human–machine control interface was implemented by employing Microsoft Foundation Classes. The communication between computer and the subsystems is through image capture card, Morphis (MOR/2VD/84) from Matrox (Quebec, Canada), AD/DA cards, PCI-1727U and PCI-1710HG from Advantech (Kaohsiung, Taiwan), and communication interface by RS-232.

3. Autofocusing Operations

A visual servo system with autofocusing operations is an important setup on gripping moving particle. The present image-based autofocusing operation is the first to focus the moving particle and gripper tips, respectively, by evaluating the autofocusing function at different in-focusing positions. Then, the in-focusing positions are utilized to estimate the depth for aligning the gripper tips with moving micro particle in the same focus plane.

Regarding the image-based autofocusing operation, the focus scheme involves searching strategy and evaluating focus measure to locate the focus position. The searching strategy usually includes Global search, Binary search, and Rule-based search [25,34,35]. The selection consideration of searching strategy depends on the searching range, speed, accuracy, and robustness. In the digital searching process, a discrete focus measure which obtains the maximum value in the focusing range is deemed as the best focused image. A discrete focus measure to be formulated is expected to have the properties

of the unbiased distribution, clear single apex, high sensitivity, good anti-noise capability, and rapid computation.

In the past, many discrete focus functions have been proposed in the literature. By summing up the different classification of focus functions proposed by Groen et al. [27], Sun et al. [30], and Xie et al. [31], the discrete focus functions in general can be categorized as Derivative-based measure [36–40], Statistics-based measure [27,39,41,42], Histogram-based measure [28], Gray level magnitude-based measure [26,27,29], and Wavelet-based measure [31,43]. In the present focusing and gripping applications, a new focus measure was proposed and the focusing performance was tested and compared with the widely-used measures.

3.1. Wavelet-Entropy Focusing Function

In an image, the diversity of pixel grey level of a focused image is usually greater than that of a defocused one. From the viewpoint of information science, a focusing process can be considered as changes of distribution content of an image. Since an entropy function is a measure of diversity as information content, it may be selected and utilized as a focusing function. Firestone et al. (1991) [28] proposed a histogram-measure Shannon entropy, which is formulated by utilizing the probability of grey level occurred in an image, as a focusing function. The focusing performance of histogram-measure entropy had been tested and compared with other focusing functions. From various comparative studies, the performance of histogram-measure entropy is poor [30,31]. For defining a focus function, it may not be appropriate to formulate entropy through grey level in space domain.

Beyond the information viewpoint on focusing process, a focusing process can be considered as changes of spatial energy content of an image. Application of wavelet energy analysis for defining focus function was proposed by Xie et al. (2007) [31]. The formulation of focus function was to utilize spatial image energy through wavelet coefficients since the orthogonal discrete wavelet transform preserves the contents of image energy. When focusing the image, the energy of high frequency will increase while the energy of low frequency will decrease. In the comparison of four focus measures by Histogram Entropy, Energy Laplace, Normalized Variance, and Wavelet Algorithm, testing results revealed that M_w, which was formulated as a ratio between high frequency energy and low frequency energy, provided the best performance.

The present focusing operation involved two different scales of objects, gripper tips and micro particles, which is different from the aforementioned focusing images under tests. Considering this, a new measure to integrate both information and energy contents in an image may be better utilized for the present applications. First, the Shannon entropy function, which can measure the contents of average information of objects, may be eligible. In the entropy formulation, the discrete Shannon entropy is defined as [28]

$$E_{1D}(X) = -\sum_{i=1}^{n} p_i \ln(p_i) \tag{1}$$

where X is a random variable with n outcomes $\{x_i: i = 1, \ldots, n\}$ and p_i is the probability of outcome x_i. Next, the probability distribution of signal energy relative to the total signal energy can be expressed:

$$p_i = |x_i|^2 / \|X\|^2 \tag{2}$$

where x_i is a series of wavelet coefficients and $\|X\|^2$ is the l^2-norm of $X = \{x_i: i = 1, \ldots, n\}$. The functional in Equation (1) utilizing p_i of Equation (2) in Shannon entropy is non-additive. A one-dimensional additive Shannon entropy may be given as

$$E_{1D}(X) = -\sum_{i=1}^{n} x_i^2 \ln(x_i^2), x_i \neq 0 \tag{3}$$

Here, it is noted that the effect of $\|X\|^2$ is only to shift a function along vertical axis and scale the magnitude of function. Thus, the additive Shannon entropy preserves the function property of Shannon entropy. However, the additive Shannon entropy gives negative value while the Shannon entropy gives positive value.

By following the entropy formulation of Equation (3) and introducing the wavelet energy, the spatial wavelet entropy for a specific wavelet domain may be defined as

$$E^l_{(x,y)\in D} = -\sum_{x=1}^{M}\sum_{y=1}^{N}\left|C^l_{x,y}\right|^2 \log(\left|C^l_{x,y}\right|^2), C^l_{x,y} \neq 0 \tag{4}$$

where D is a wavelet domain on an image and $C^l_{x,y}$ are wavelet coefficients at level l. The image entropy can be expressed as a summation of the entropies contributed by different regions in the wavelet decomposition. By utilizing the mother-wavelet filter of high pass (H) and low pass (L) to an image, the resultant image is divided into four sub-images: LL, HL, LH, snd HH. For level-1 decomposition, the total entropy can be expressed as

$$E^1_{(x,y)\in D} = E^1_{(x,y)\in HH} + E^1_{(x,y)\in HL} + E^1_{(x,y)\in LH} + E^1_{(x,y)\in LL} \tag{5}$$

Since the second moment of wavelet coefficients in the low frequency component will be slightly decreased in the focusing process, this component does not contribute useful focusing information. Consequently, the component of LL in Equation (5) can be ignored and an autofocusing function in the level-1 decomposition, including total high frequency components, is proposed:

$$E_{THF1} = E^1_{(x,y)\in HH} + E^1_{(x,y)\in HL} + E^1_{(x,y)\in LH} \tag{6}$$

For higher level wavelet decomposition, an autofocusing function in the level-2 decomposition can be considered. In considering the issue of real-time computation, only three terms in wavelet decomposition are retained. The contribution of focusing information in the level-2 decomposition is to retain the higher and lower frequency components than those components in the level-1 for obtaining higher contrast between the high frequency and low frequency contents. From Equation (6), an autofocusing function to include significant partial high frequency components in level 2 is proposed:

$$E_{PHF2} = E^2_{(x,y)\in HLHL} + E^2_{(x,y)\in LHLH} + E^2_{(x,y)\in HHHH} \tag{7}$$

The present wavelet-entropy focusing functions are proposed in Equations (6) and (7). The performance of E_{THF1} and E_{PHF2} in the gripping applications needs to be tested.

3.2. Experimental Test and Comparison with Other Focusing Functions

The performance of the proposed autofocusing function on the moving particle was compared with both common focusing functions and wavelet-based focusing functions. In the experimental tests, the particles were made of Polystyrene with density 1.05 g/mL and size 30–50 μm. The microparticles were suspended in water over a long duration. In the experimental tests, the microgripper tips were positioned in the same focus plane as that of microparticles in advance. In utilizing the Autofocusing stage for focusing tests, the block diagram shown in Figure 4 was operated in open loop. The functional block of the Procedure of autofocusing operation in PC was to implement various focusing functions in capturing particle images at different Z positions. The size of particles image was 640 × 480 pixels. The image was captured on every one cyclic rotation of step motor. A total of 29 images were collected. In the sequence of 29 images, Images 1 to 12 were captured in focusing process while Images 13 to 29 were captured in defocusing process. The 29 images in sequence are listed as shown in Figure 4, where the best focusing image is found at image 12.

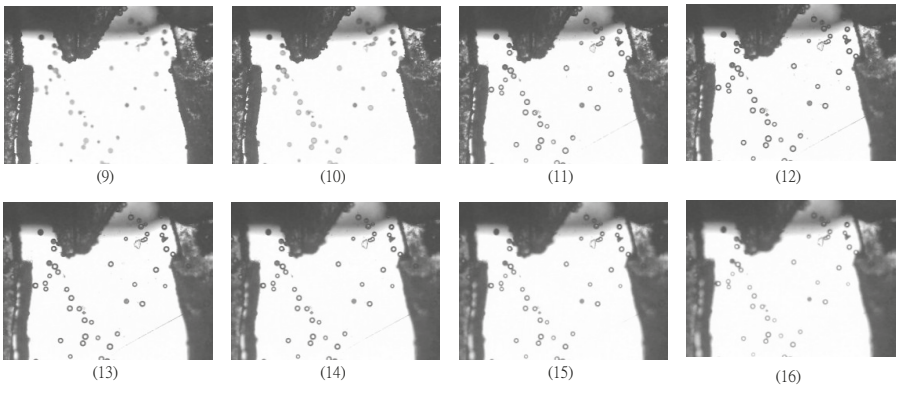


Figure 4. A sequence of eight images from total 29 images are displayed to illustrate the test of autofocusing functions. Images (**1**)–(**12**) are captured in the focusing process while Images (**13**)–(**29**) are captured in the defocusing process. Note that the best focusing image is at Image (**12**).

For the evaluation of noise effect on focusing functions, the relative strength of the added noise is measured by employing *SN* ratio in dB

$$SNR = 10\log_{10}\frac{\frac{1}{M\times N}\sum_{i=1}^{M}\sum_{j=1}^{N}(IM)_{ij}^{2}}{\frac{1}{M\times N}\sum_{i=1}^{M}\sum_{j=1}^{N}((IM)_{ij}-(IMZ)_{ij})^{2}} \quad (8)$$

where $M \times N$ is the size of original image, IM is the grey level of the original image, and IMZ is the grey level of the original image with added noise. In the evaluation of noise effect, a Gaussian white noise with variance 0.0008 was added to each image. The SNR of 29 test images was calculated to give 26 ± 0.2 dB.

3.2.1. Comparison with Common Focusing Functions

The focusing performances of the proposed E_{THF1} and E_{PHF2} are compared with the common focusing functions: Normalized variance function, Entropy function, Energy Laplace function, Tenenbaum Gradient function, and Sum Modulus Difference (SMD) function. In the wavelet-based algorithms, the mother wavelet was Haar function, i.e., db2 or Daubechies 2. In comparing performance, the evaluated results of autofocusing functions were normalized, as shown in Figure 5. Figure 5 reveals that all focusing functions give the maximum functional values at the same Z position. The Z position at 12 is a correct in-focus position. However, the autofocusing functions E_{THF1} and E_{PHF2} give sharper curves than those of other autofocusing functions. In addition, the performance of the double decomposition E_{PHF2} is somewhat better than that of single decomposition E_{THF1}.

3.2.2. Comparison of Wavelet-Based Focusing Functions

The autofocusing performance by wavelet-entropy function E_{THF1} and E_{PHF2} was compared with the wavelet-based autofocusing functions by M_w, M1, M2 and M3. Here, the autofocusing functions M1, M2 and M3 correspond to W1, W2, and W3 in [30], respectively. In the comparisons of performance, the evaluated results of autofocusing functions were normalized, as displayed in Figure 6. Figure 6 reveals that all focusing functions give the maximum functional values at the same correct Z position. Among the wavelet-based focusing functions, the present double decomposition E_{PHF2} gives the best performance.

The aforementioned comparison results reveal that the wavelet-entropy function E_{PHF2} gives the best performance in focusing both microparticles and gripper tips. Therefore, the E_{PHF2} is selected and employed for the depth estimation to align microgripper tips and moving particle in the same focus plane.

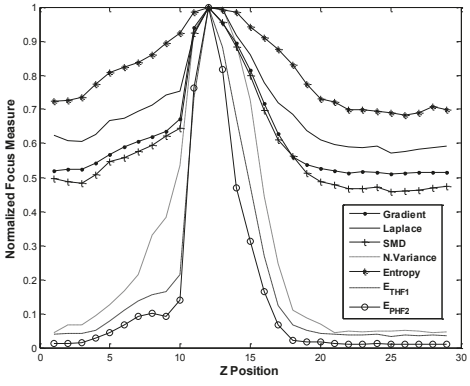


Figure 5. Comparisons of focus measures by wavelet-entropy E_{THF1}, E_{PHF2} and other common focus functions (SNR of test image = 26 $\pm$ 0.2 dB).

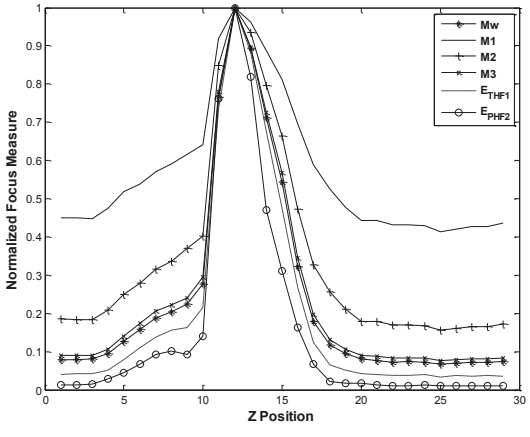


Figure 6. Comparisons of focus measures by wavelet-entropy E_{THF1}, E_{PHF2} and other wavelet-based functions (SNR of test image = 26 $\pm$ 0.2 dB).

3.3. Peak Position Identification and Depth Estimation in Visual Servo

3.3.1. Peak Position Identification

In the image-based microscopic autofocus, Global searching strategy was selected for identifying the peak position of focus measure. The image was captured on every one cyclic rotation of step motor. A total of 20 images were collected. The result of E_{PHF2} at different Z positions is shown in Figure 7. Figure 7 shows that there are several local maxima in the focusing function. The highest peak at position 16 refers to the image of the in-focus position of micro particles. The next highest peak at position 3 refers to the image of the in-focus position of gripper tips. To obtain the correct identification of the two local maxima in different Z positions, a template of gripper tips, which was

not changed in the process of Global searching, was utilized to distinguish two focusing positions by evaluating the matching score between images. Figure 8 displays the matching score in the process of Global searching. In Figure 8, it is observed that the highest score at position 3 is obviously the in-focus position of gripper tips. The lowest score at position 16 is the in-focus position of microparticles.

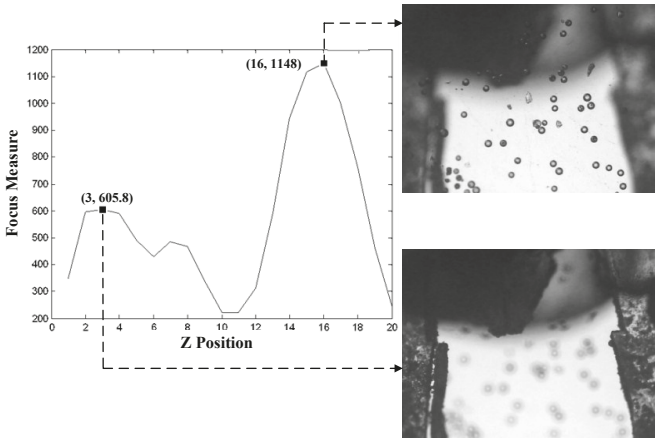


Figure 7. Focus measure of E_{PHF2} in focusing gripper tips and particles.

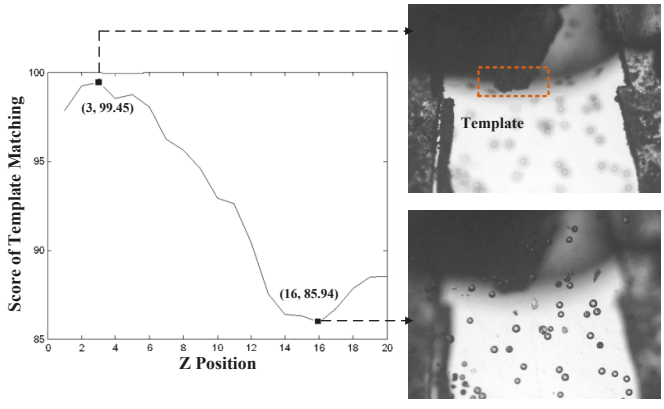


Figure 8. Score of template matching by employing the template of gripper tips.

3.3.2. Depth Estimation

After identifying the in-focus positions of gripper tips and microparticles, the depth between these two positions in Z axis can be evaluated by:

$$Depth = \left| P_{particle} - P_{gripper} \right| \times M_{object_len} \quad (9)$$

where $P_{particle}$ and $P_{gripper}$ are the Z positions of particles and gripper tip, respectively, and M_{object_len} is the moving distance of object lens by stepping motor in one revolution. Here, M_{object_len} = 22.20 μm/rev. is calculated by the multiplication of gain of step motor, gear ratio, and microscope focus from Figure 3. $P_{particle}$ = 16 rev. and $P_{gripper}$ = 3 rev. are obtained from Figure 8. Consequently, the depth between two

focusing plane is calculated to give *Depth* = 288.6 μm. In the servo loop, the depth is calculated in real time for positioning the gripper tips to the same image plane of particles.

The visual servo of autofocusing operation is given by Figure 3. In Figure 3, the functional block of the Procedure of autofocusing operation in PC is to implement the autofocusing operation given in Figure 9. Here, it is noted that the position of gripper tips in *Z* axis will be moved to the same focusing image plane as that of particles before carrying out the next process of tracking particle.

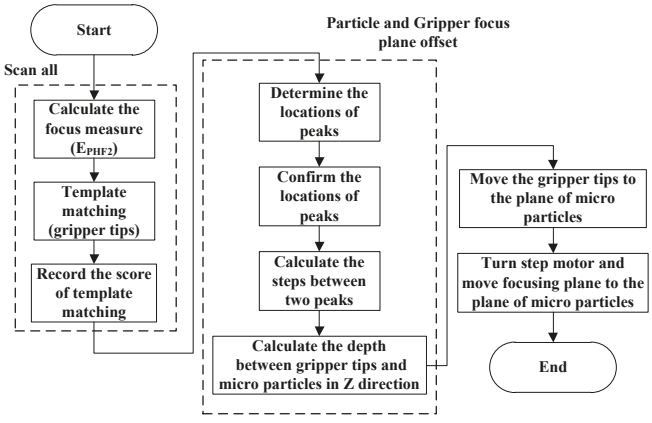


Figure 9. Flowchart of autofocusing operation to align microgripper tips and moving particles in the same focus plane (coarse adjustment).

4. Particle Tracking Process

In utilizing the image-based digital PTV for tracking particle, the first step is to identify and label particles. The next step is to estimate the centroid coordinates of particles. In the last step, the positions of each particle between two consecutive time step images are matched. In considering the requirements of real-time and robust tracking operation, a two-frame scheme by Polar Coordinate System Similarity (PCSS) [44], which is insensitive to experimental parameters and applicable to flows subjected to strong rotation, is first selected and considered for the present applications.

4.1. PCSS Algorithm

The essence of PCSS for finding the matched particles in tracking operation is employing a similarity measure to estimate the changes of both distance and angle of surrounding particles relative to the other particles at two consecutive image frames. The PCSS algorithm is implemented by the following steps:

Step 1. Set an interrogation pattern I centered on the object particle i with radius of R. In the pattern I, the other particles are i_m, where $m = 1, \ldots, M$.

Step 2. Establish a polar coordinate system with the identified centroid of the particle i as the pole. The relative positions of i_m to i is obtained by their polar radii r_{im} and angles θ_{im}.

Step 3. Repeat Steps 1 and 2 for another interrogation pattern J centered on the object particle j with radius of R. In the pattern J, the other particles are j_n, where $n = 1, \ldots, N$, and the relative positions of j_n to j is obtained by their polar radii r_{jn} and angles θ_{jn}.

Step 4. Define a similarity coefficient between patterns I and J as

$$S_{ij} = \sum_{n=1}^{N} \sum_{m=1}^{M} H(\varepsilon_r - |r_{im} - r_{jn}|, \varepsilon_\theta - |\theta_{im} - \theta_{jn}|) \quad (10)$$

$$H(x,y) = \begin{cases} 1, & x > 0,\ y > 0 \\ 0, & otherwise \end{cases} \tag{11}$$

where $H(x,y)$ is a step function. ε_r and ε_θ are thresholds of the polar radius and angle, respectively.

Step 5. Calculate the S_{ij} for all candidate particles in the patterns I and J, and find the matched candidate particle i which gives the maximum value of S_{ij}.

The application of PCSS in tracking particles requires the condition of small velocity change in flow field [32]. However, in considering the present application of tracking micron particle, the particle may fluctuate in Brownian motion which causes low value of S_{ij} in tracking algorithm. Therefore, an enhanced PCSS is proposed to incorporate template matching in the tracking algorithm when the value of S_{ij} is not well above a threshold value ρ.

4.2. Template Matching

The representing particle position in fluctuating movement is obtained through template matching technique. By utilizing zero-mean normalized cross-correlation, the correlation between a template image $w(i,j)$ of size $K \times L$ and an input image $f(x,y)$ of size $M \times N$ can be expressed as

$$\gamma(x,y) = \frac{\sum_{i=0}^{L-1}\sum_{j=0}^{K-1}(w(i,j)-\overline{w})(f_c(x+i,y+j)-\overline{f}_c(x,y))}{\left\{\sum_{i=0}^{L-1}\sum_{j=0}^{K-1}(w(i,j)-\overline{w})^2\sum_{i=0}^{L-1}\sum_{j=0}^{K-1}(f_c(x+i,y+j)-\overline{f}_c(x,y))^2\right\}^{1/2}} \tag{12}$$

where $\overline{w}$ and $\overline{f}_c$ are the mean of w and f_c, respectively, and the f_c is a subimage of the same size of template. The best match between the template and image is given by the maximum correlation:

$$\gamma(x',y') = \max_{x,y}\{\gamma(x,y)x = 1 \sim M, y = 1 \sim N\} \tag{13}$$

The enhanced PCSS algorithm combines PCSS, edge detection, and template matching in tracking particle movement. In addition, the centroid of particle is estimated by utilizing circular Hough transformation [45]. Regarding the template matching, image pyramids can be employed to enhance the computational efficiency of the correlation-based template detection. The practical operation of the enhanced PCSS algorithm for tracking particle is given by the flowchart shown in Figure 10. In Figure 10, the threshold values are assigned as $\rho = 0.7$, $\varepsilon_r \leq 5$ pixel, and $\varepsilon_\theta \leq 5^\circ$.

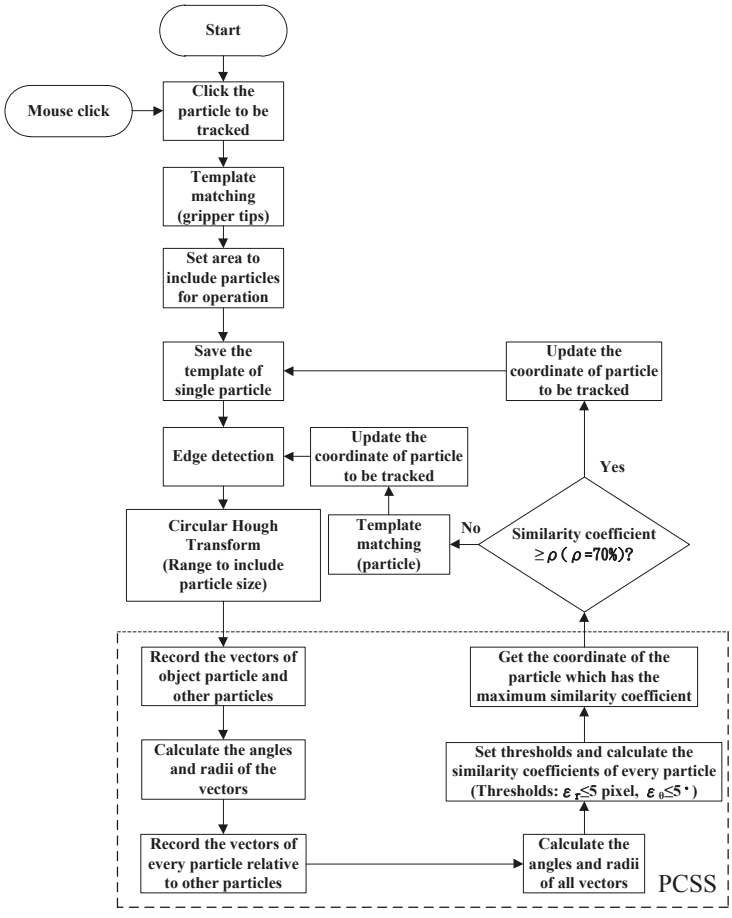


Figure 10. Enhanced PCSS for tracking micro particle on *X*-*Y* plane.

5. Gripping Microparticle Tests

A visual-servo automatic micromanipulating system was developed for gripping the moving microparticle suspended in liquid well. The performance of immersing the PU gripper tips into water to grip moving Polystyrene particle was tested. In the experimental test, the microgripper was operated with autofocusing and visual tracking functions in gripping operations. The experimental calibration gave $x_{image_calibration} = 2.494$ μm/pixel and $y_{image_calibration} = 2.491$ μm/pixel for the calibrated x, y distance per unit pixel, respectively. The size of microparticle was 30–50 μm. The microparticle was suspended in the water well. The water temperature was 25 °C. To emulate the effect of manipulating performance under flow field, the Object Platform to support the water well was moving with speed 0.6 mm/s which was orthogonal to the gripper axis. The moving speed was set through control panel.

5.1. Working Space in Gripping Operation

In the operation of gripping particle, the working space will be established. For the horizontal planner area, it is constrained by the sides of two flexible arms and the line passed through the centroid

position of gripper tips' template, as shown in Figure 11. The vertical height is constrained by the 200 μm thickness of microgripper.

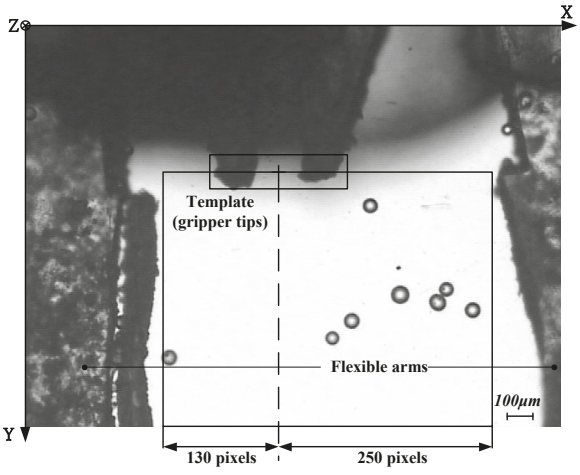


Figure 11. Horizontal planner area to constrain moving particles.

5.2. Fine Adjustment of Tracking in Z axis

Regarding the microparticle suspended in water, the focusing position of individual microparticle can change even though the coarse adjustment in the *Z*-axis focusing operation is completed. Therefore, fine adjustment of focus position is required for the accurate tracking of the specific particle moving in *Z* axis. For the fine tracking operation, at first, a planner neighborhood region of the identified particle is established:

$$area_length = 2\left(R_{\max} + V_x t_{sampling}\right) \tag{14a}$$

$$area_width = 2\left(R_{\max} + V_y t_{sampling}\right) \tag{14b}$$

where $R_{\max}$ is the maximum diameter of particle, $t_{sampling}$ is the sampling time, and V_x and V_y are the estimated particle speed moving in the *X* and *Y* direction, respectively. The center of the region is located at the centroid position of the identified particle, which is estimated by the tracking algorithm. Then, a refocusing search is initiated and the focusing measure of E_{PHF2} is recorded at each time step. The fine focusing operation will not stop until the peak value of E_{PHF2} in the next time step is little degraded. The degraded focusing measure can be detected by setting a threshold value δ:

$$\left| \frac{E_{PHF2}(t_i)_{peak} - E_{PHF2}(t_{i+1})_{peak}}{E_{PHF2}(t_i)_{peak}} \right| \leq \delta. \tag{15}$$

In the experimental tests, $R_{\max} = 20$ pixel, $V_x = V_y = 240$ pixel/s, $t_{sampling} = 0.1$ s, and $\delta = 0.2$.

5.3. Pre-Positioning and Approaching Operations

Before carrying out the operation of gripping identified particle, the microgripper will pre-position itself and then approach to the particle by visual servo. In the first pre-positioning step, the microgripper will track the particle in the horizontal *X*-*Y* plane. It will not stop until the centroid of gripper tips' template is located in a region which is close to the identified microparticle. For avoiding collision between gripper tips and particle, a non-colliding safe distance is set to 1.0 $R_{\max}$. The planner

region to pre-position the gripper is depicted in Figure 12. The gripper will stop when the following conditions are satisfied:

$$move_x(t_j) = -(gripper_x(t_j) - particle_x(t_j)) \times x_{image_calibration} \tag{16a}$$

$$\left|mov_x(t_j)\right| \leq 0.3R_{\max} \times x_{image_calibration} \tag{16b}$$

and

$$mov_y(t_j) = ((particle_y(t_j) - gripper_y(t_j))) \times y_{image_calibration} \tag{16c}$$

$$\left|mov_y(t_j) - 2R_{\max} \times y_{image_calibration}\right| \leq 0.5R_{\max} \times y_{image_calibration} \tag{16d}$$

where $particle_x(t_j)$, $particle_y(t_j)$ are the X, Y coordinates of particle position, respectively, $gripper_x(t_j)$, $gripper_y(t_j)$ are the X, Y centroid coordinates of gripper tips' template, respectively, and $mov_x(t_j)$, $mov_y(t_j)$ are the calculated distances of moving gripper tips to the particle in X, Y directions, respectively. In the second approaching step, the centroid of gripper tips' template will move to approach the particle. By employing visual servo, the input command for the planner movement of microgripper in X and Y directions can be calculated from Equations (16a) and (16c), respectively. If the particle is not accurately tracked by the gripper, the second approaching step can be interrupted by sending operator's command.

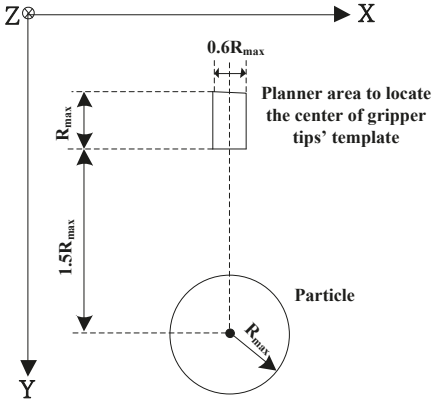


Figure 12. Planner area to locate the gripper by the center of gripper tips' template in pre-positioning operation.

5.4. Gripping and Releasing Operations

For gripping microparticle, the gripper tip is continuously moving to approach the microparticle. It will not move until the position of particle is located near the centroid of gripper tips' template. Then, a gripping command is initiated to close and grip the particle. After gripping particle, the gripper will move upward in $-Z$ direction and then hold the particle to 400 μm away in both X and Y directions. Finally, the gripper tips will move to approach the bottom of well and then the gripper will open for releasing microparticle. To avoid adhesion of the particle to the gripper, the microgripper will move fast along $-Y$ for flushing the particle away.

In the gripping microparticle tests, the operational procedures were: (1) initiating the movement of gripper and capturing gripper tips' template; (2) coarse adjustment; (3) identifying particle by mouse click; (4) fine adjustment; (5) tracking microparticle on X-Y plane; (6) pre-positioning gripper; (7) gripper approaching particle; (8) gripping, transporting, and releasing particle; and (9) homing of gripper. In the initial operation, the microgripper was moved to the position of water surface and the template of gripper tips was captured. During the operational process, the Stop Grasping Command

was waited for operator to input if the identified particle was not well tracked by the fine adjustment of focus. The detail operational flowchart is shown in Figure 13. In the performance tests, the images of tracking and gripping identified particle in the 3-D space were captured. The six captured images are displayed in Figure 14. Figure 14(1) shows the initial image in the beginning of global searching for coarse adjustment. Figure 14(2) shows the image of finishing autofocus adjustment. Figure 14(3) displays the image of clicking mouse on the identified particle. Figure 14(4) displays an image in the process of approaching particle before gripping operation. Figure 14(5) displays an image in the process of gripping particle. Figure 14(6) displays an image in the process of releasing particle after the completion of gripping and transporting particle. For the experimental tests, the time expense from the beginning of the global search for coarse adjustment to the actual gripping of the particle was less than 50 s. The successful gripping rate of the present apparatus was 8/10 when the object platform was moving in 0.6 mm/s. The failure was mainly due to the adhesion of micro particles after fine adjustment which caused the inaccurate estimation of particle's coordinates. As a result, the proposed apparatus can perform better when the particle concentration is somewhat low.

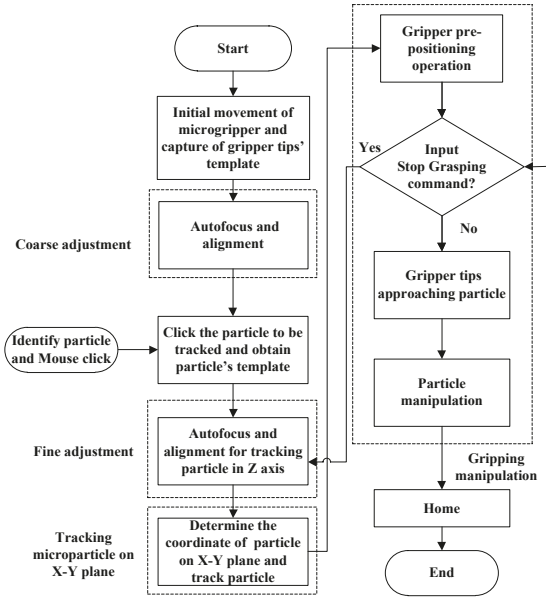


Figure 13. The processes of tracking and gripping particle moving in 3-D space.

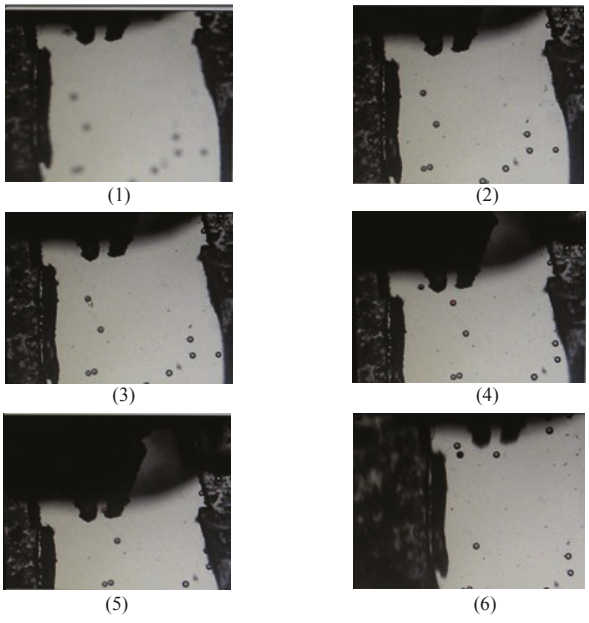


Figure 14. Results of tracking and gripping particle in 3-D space: (**1**) beginning of global searching for coarse adjustment; (**2**) finishing autofocus adjustment; (**3**) identifying particle by clicking mouse; (**4**) approaching particle; (**5**) gripping particle; and (**6**) releasing particle after completion of gripping and transporting particle.

6. Conclusions

An innovative visual-servo automatic micromanipulating system has been developed for gripping the moving microparticle suspended in liquid well. By using an innovative piezoelectric actuated PU micro gripper with flexible arms, a microparticle can be tracked and gripped in a moving work space. A new wavelet-based focus function E_{PHF2}, which revealed the best performance among the surveyed focus functions, was proposed and utilized to estimate the depth for the alignment of microgripper tips and moving particle in the same focus plane. An enhanced PCSS algorithm incorporated with template matching, edge detection method, and circular Hough Transform was implemented for tracking particles under fluctuating and drifting flow. In the performance tests, the images of experimental results in the processes of tracking, gripping, transporting, and releasing 30–50 μm Polystyrene particle in 25 °C water were captured and displayed. For the experimental tests, the time expense from the beginning of the global search for coarse adjustment to the actual gripping of the particle was less than 50 s. The successful gripping rate of the present apparatus was 8/10 when the object platform was moving in 0.6 mm/s. The failure was mainly due to the adhesion of micro particles after fine adjustment which caused the inaccurate estimation of particle's coordinates. As a result, the proposed apparatus can perform better when the particle concentration is somewhat low.

Author Contributions: Methodology, R.J.C.; Writing-Review & Editing, R.J.C.; Software, Y.C.C.; Investigation, R.J.C. and Y.C.C.

Acknowledgments: The authors would like to thank the Ministry of Science and Technology (MOST) for the partial support provided under contract No. MOST 105-2221-E-006-081.

Conflicts of Interest: The authors declare no conflicts of interest.

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Innovative Silicon Microgrippers for Biomedical Applications: Design, Mechanical Simulation and Evaluation of Protein Fouling

Cristina Potrich [1,2,*], Lorenzo Lunelli [1,2], Alvise Bagolini [3], Pierluigi Bellutti [3], Cecilia Pederzolli [1], Matteo Verotti [4,5] and Nicola Pio Belfiore [6]

1 Laboratory of Biomarker Studies and Structure Analysis for Health, Fondazione Bruno Kessler, Trento I-38123, Italy; lunelli@fbk.eu (L.L.); pederzo@fbk.eu (C.P.)
2 CNR- Institute of Biophysics, Unit at Trento, Trento I-38123, Italy
3 Micro Nano Fabrication and Characterization Facility, Fondazione Bruno Kessler, Trento I-38123, Italy; bagolini@fbk.eu (A.B.); bellutti@fbk.eu (P.B.)
4 Department of Industrial Engineering, University of Trento, Trento 38123, Italy; matteo.verotti@unitn.it
5 Mechatronics Prototyping Facility, Trentino Sviluppo S.p.A., Rovereto 38068, Italy
6 Department of Engineering, Università degli Studi di Roma Tre, via della Vasca Navale 79, Roma 00146, Italy; nicolapio.belfiore@uniroma3.it
* Correspondence: cpotrich@fbk.eu; Tel.: +39-0461-314605

Received: 8 March 2018; Accepted: 21 March 2018; Published: 24 March 2018

Abstract: The demand of miniaturized, accurate and robust micro-tools for minimally invasive surgery or in general for micro-manipulation, has grown tremendously in recent years. To meet this need, a new-concept comb-driven microgripper was designed and fabricated. Two microgripper prototypes differing for both the number of links and the number of conjugate surface flexure hinges are presented. Their design takes advantage of an innovative concept based on the pseudo-rigid body model, while the study of microgripper mechanical potentialities in different configurations is supported by finite elements' simulations. These microgrippers, realized by the deep reactive-ion etching technology, are intended as micro-tools for tissue or cell manipulation and for minimally invasive surgery; therefore, their biocompatibility in terms of protein fouling was assessed. Serum albumin dissolved in phosphate buffer was selected to mimic the physiological environment and its adsorption on microgrippers was measured. The presented microgrippers demonstrated having great potential as biomedical tools, showing a modest propensity to adsorb proteins, independently from the protein concentration and time of incubation.

Keywords: microgripper; biocompatibility; mini-invasive surgery; protein fouling; cell manipulation

1. Introduction

The demand for micro-manipulation in biomedical applications increased during the last decades. Hundreds of different micro-systems have indeed been reported in the literature [1,2], with different operational strategies [3] and being mainly dedicated to medical or biological applications. Among others, an untethered mobile microgripper controlled by an external electromagnetic coil system has been reported [4], where the manipulation of different microgels was successfully demonstrated with a pick-and-place robotic heterogeneous 3D assembly technique. A few more examples of microgrippers are present in the literature, piezoelectric [5,6] or thermally [7,8] or even magnetically actuated [8–10]. All these systems, however, present quite large dimensions and are not fully tested for working in a physiological environment, as required for biomedical applications.

The development of new microsystems and, in particular, silicon microgrippers for handling cells or tissues at the microscale was therefore pursued in this paper. Since such manipulators have to handle